Deep Learning

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Escuela de Ciencias Aplicadas e Ingeniería

Introducción

https://github.com/jdmartinev/SI3014---Redes-Neuronales-y-Aprendizaje-Profundo/blob/main/README.md

Artificial Intelligence, deep learning, machine learning—whatever you're doing if you don't understand it—learn it. Because otherwise you're going to be a dinosaur within 3 years.

-Mark Cuban

Contenido y Evaluación

➢ Modulo I → Fundamentos del Deep Learning (DL)

- ¿Qué es Deep Learning?
- Introducción a PyTorch
- Tensores y operaciones básicas
- Álgebra Lineal para Deep Learning (Producto punto y producto matricial)
- Fundamentos de Redes Neuronales: Neurona artificial, Perceptrón, Funciones de activación (ReLU, Sigmoid, Tanh, Softmax), Funciones de pérdida
- Bases de datos: Organización (Train/Validation/Test), Dataset y DataLoader
- Optimización de entrenamiento: Descenso del gradiente, SGD, Adam
- Primer modelo de regresión en PyTorch

Eventos evaluativos

- Parcial I 20% - Semana 4
- Laboratorio I 10% - Semana 5

Contenido y Evaluación

- **Modulo II → Modelos Convolucionales**
  - Convolutional Neural Networks (CNN)
  - AutoEncoders
  - U-Net Neural Networks
  - Optimizadores
  - Overfitting y underfitting.
  - Hiperparámetros
  - Grid search vs Random search
  - Early stopping
  - PCA vs Autoencoders

Eventos evaluativos

- Quiz I 10% - Semana 7
- Laboratorio II 10% - Semana 8

Contenido y Evaluación

➢ Modulo III → Modelos Generativos y de Atención

- VAE
- GANs y cGANs
- Motivación de la atención
- Atención vs RNN
- Self-AttentionTransformers (visión general)
- Encoder–Decoder
- Casos de uso: texto, imágenes, señales
- Limitaciones y costos computacionales

Eventos evaluativos

➢ Quiz II 10% - Semana 12

Contenido y Evaluación

➢ Modulo IV → Modelos Secuenciales

- ➢ Redes Neuronales Recurrentes (Recurrent Neural Networks, RNN)
- ➢ Problemas del Desvanecimiento y Explosión del Gradiente (Vanishing and Exploding Gradient Problems)
- ➢ Long Short-Term Memory (LSTM)
- ➢ Gated Recurrent Unit (GRU)
- ➢ Modelado de Series Temporales (Time Series Forecasting)
- ➢ Fundamentos del Procesamiento del Lenguaje Natural (Natural Language Processing, NLP)

Eventos evaluativos

- ➢ Quiz III 10% - Semana 15
- ➢ Laboratorio III 10% - Semana 16

Panorama global

Evento Evaluativo	Porcentaje	Semana
Parcial I	20 %	Semana 4
Laboratorio I	10 %	Semana 5
Quiz I	10 %	Semana 7
Laboratorio II	10 %	Semana 8
Quiz II	10 %	Semana 12
Quiz III	10 %	Semana 15
Laboratorio III	10 %	Semana 16
Proyecto Final	20 %	Semana 18

Nota: Los laboratorios y el proyecto final se desarrollarán en grupos de máximo tres estudiantes. Los grupos se conformarán durante la primera clase y se mantendrán **sin cambios** durante todo el semestre.

Reglas generales

Asistencia

La inasistencia se penalizará de la siguiente manera sobre la nota del proyecto final:

Inasistencia de 12 horas: penalización del **20 %**.

Inasistencia de 9 horas: penalización del **10 %**.

Inasistencia de 6 horas: penalización del **5 %**.

Hasta 3 horas de inasistencia: se considerarán como margen de tolerancia y no tendrán penalización.

Compromisos de los estudiantes

Realizar y entregar las actividades en las fechas definidas.

Aportar significativamente al equipo de trabajo.

Estar atento a sus calificaciones durante todo el semestre.

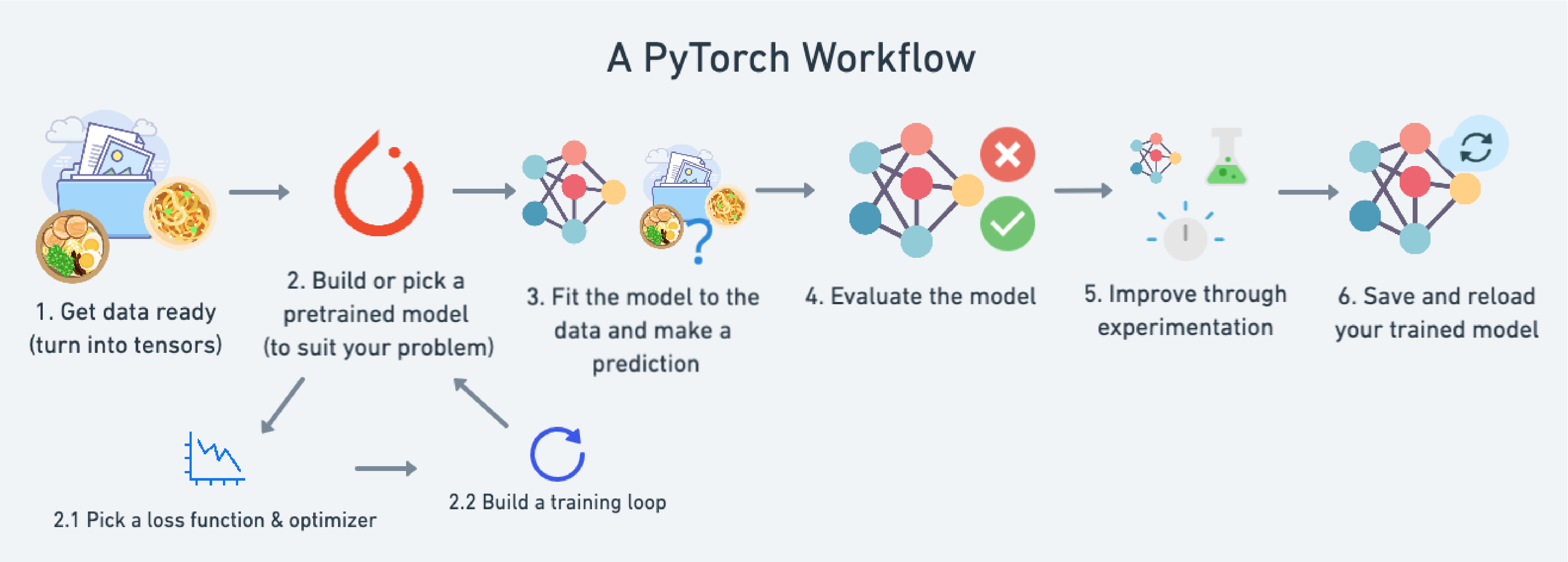
Leer y seguir los lineamientos del reglamento, el cual está disponible en https://www.eafit.edu.co/institucional/reglamentos/Documents/reglamento-academico-pregrado.pdf.

Información Importante

Actividad	Fecha	Observaciones
Inicio y terminación de clases	Julio 13 / Noviembre 07	Del 09 al 21 de noviembre se pueden utilizar para la realización de evaluaciones finales.
Semana de receso	Octubre 05 / Octubre 11	No clases – No eventos evaluativos.
Fecha límite para solicitar la suspensión de semestre	Octubre 02	Consultar regularmente las solicitudes de cancelación.
Fecha límite para reportar 70%	Noviembre 04	En EAFIT Interactiva.
Fecha límite de cancelación de materias	Noviembre 06	Consultar regularmente las solicitudes de cancelación.
Fecha límite para reportar 100%	Noviembre 24	En EAFIT Interactiva.

Framework

En Deep Learning no escribimos programas lineales; trabajamos con un *workflow experimental*
PyTorch proporciona la estructura para definir, entrenar y evaluar modelos de forma iterativa.



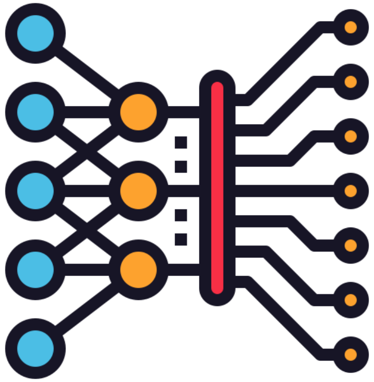
Framework

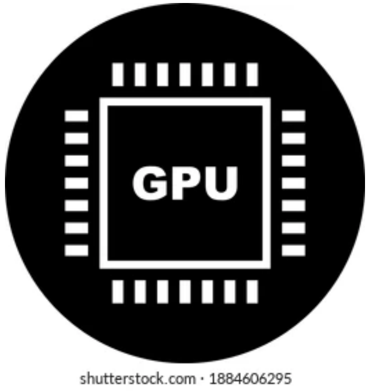
- **PyTorch** es un framework de **cómputo científico de código abierto**, basado en Python, diseñado para realizar **operaciones numéricas de alto rendimiento** y para el desarrollo de **modelos de Deep Learning**.

- Es ampliamente utilizado tanto en **investigación** como en **aplicaciones industriales**, debido a su flexibilidad, eficiencia y facilidad de uso.

Público Objetivo

- Usuarios de cómputo numérico: Funciona como un reemplazo de NumPy que permite ejecutar operaciones sobre tensores acelerados por GPU.
- Investigadores y desarrolladores en Deep Learning: Proporciona una plataforma flexible y eficiente para diseñar, entrenar y experimentar con redes neuronales.





¿Qué necesitas para el curso?

- Python – IDE

jupyter

ANACONDA

Visual Studio Code

Lightning AI

- Python libraries

matplotlib

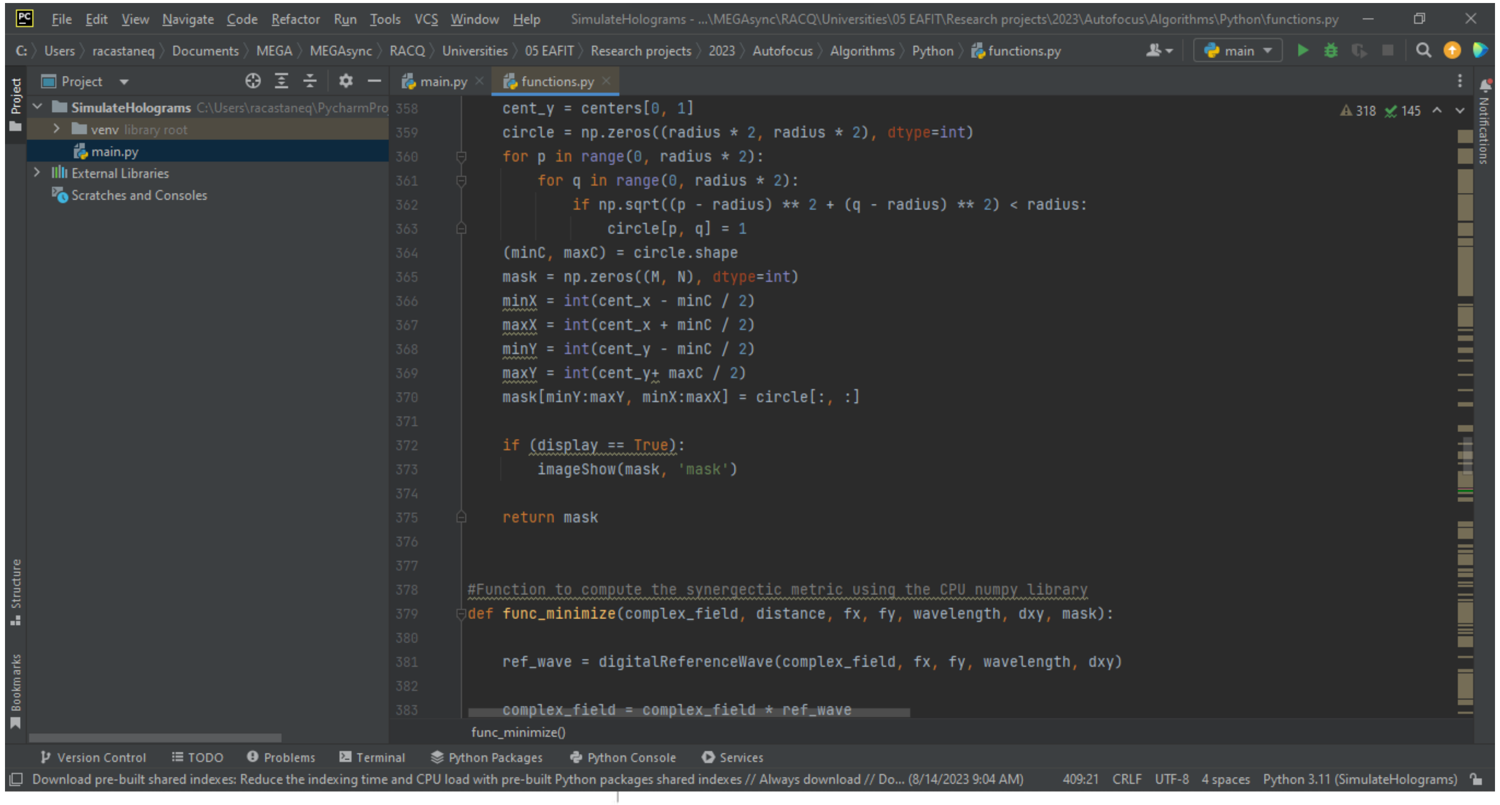
pandas

scikit learn

seaborn

Pycharm

PyCharm es un Entorno de Desarrollo Integrado (IDE) diseñado específicamente para la programación en Python. Es desarrollado por JetBrains, una empresa reconocida por crear herramientas de desarrollo de software potentes y fáciles de usar. PyCharm ofrece una amplia variedad de funciones para mejorar la experiencia de desarrollo en Python y aumentar la productividad. Algunas de sus características principales incluyen:



- Editor de código
- Análisis inteligente de código
- Debugger (Depurar)
- Gestión de Proyectos

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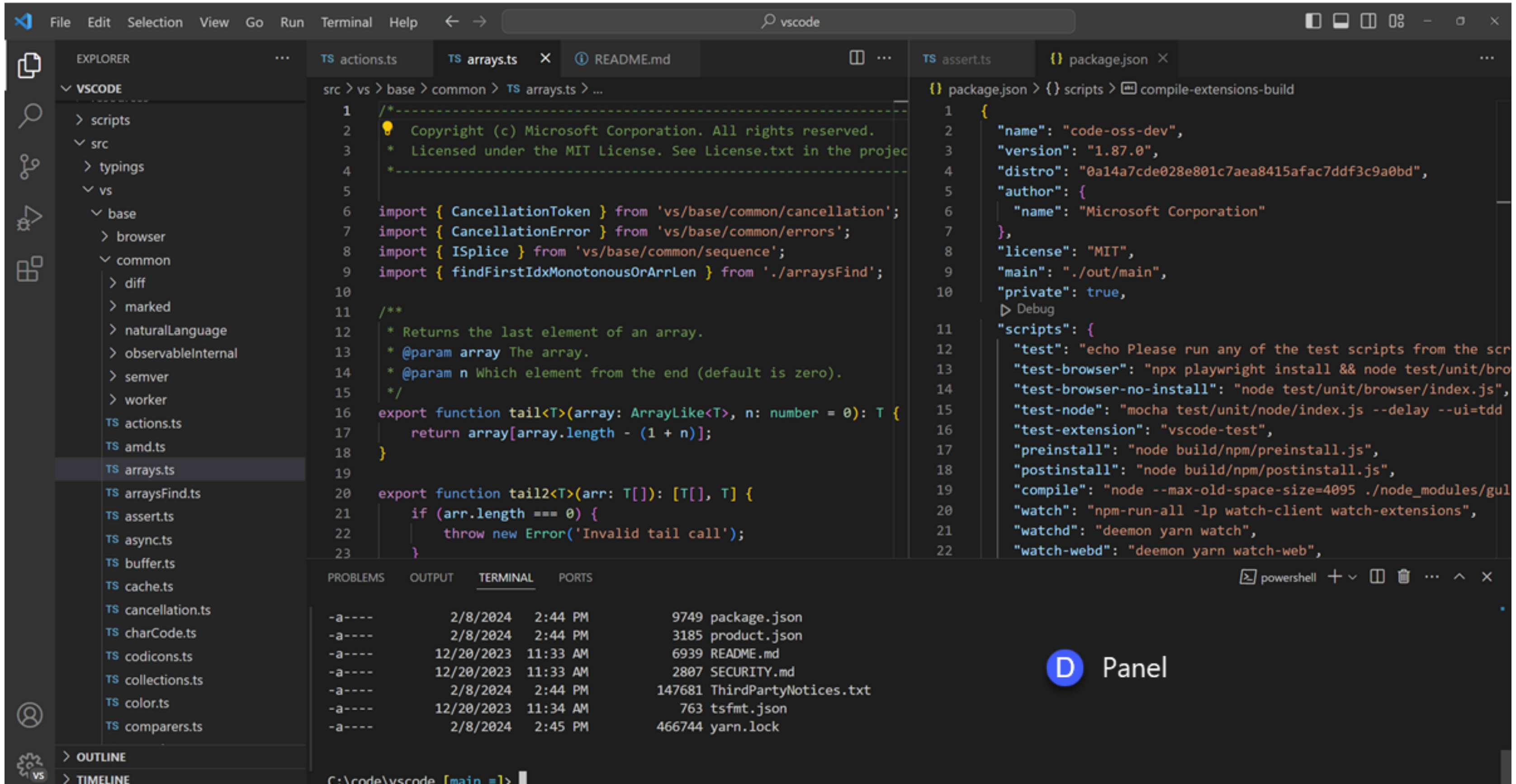
Visual Studio Code

Visual Studio Code es un editor de código fuente multiplataforma, ligero y altamente configurable, utilizado para el desarrollo de software en múltiples lenguajes de programación, incluido Python. Es desarrollado por Microsoft y se destaca por su flexibilidad, rapidez y amplio ecosistema de extensiones. Visual Studio Code ofrece herramientas que facilitan la escritura, depuración y organización del código, permitiendo al desarrollador personalizar el entorno de trabajo según sus necesidades y mejorar su productividad.

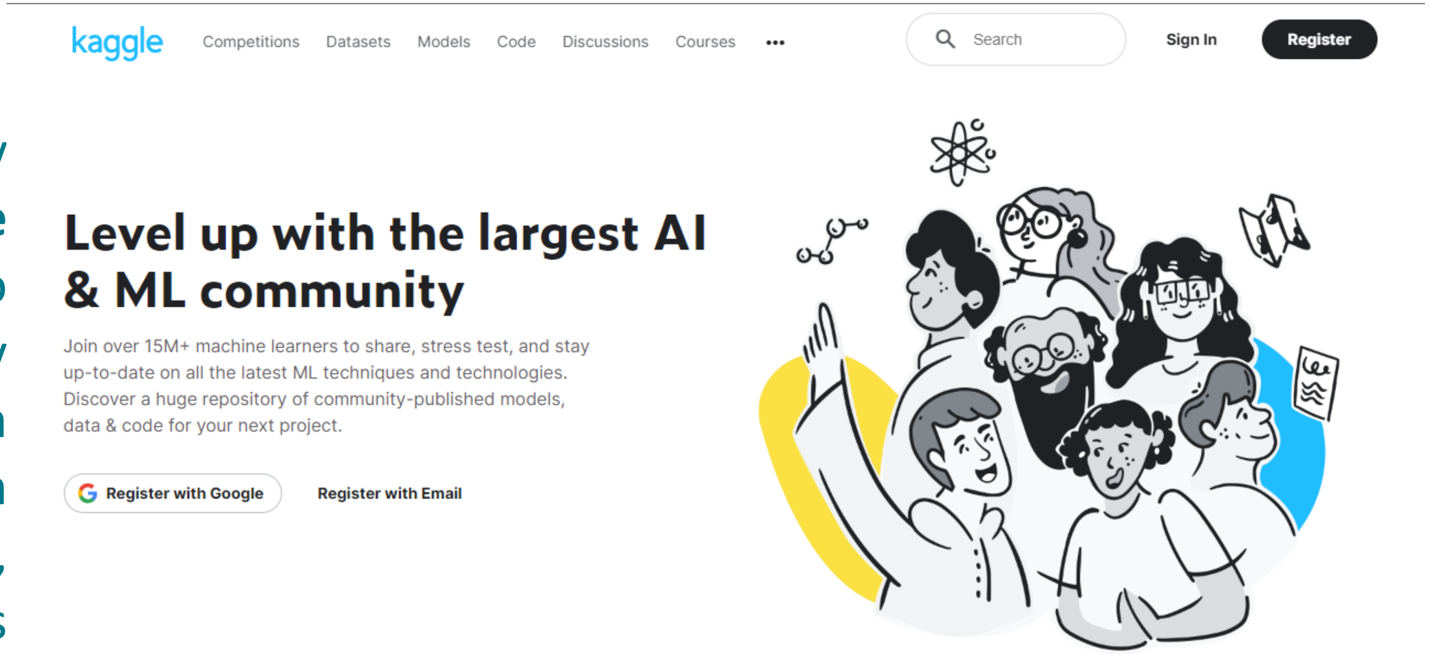


Fuentes – Bases de datos

kaggle

[Link]

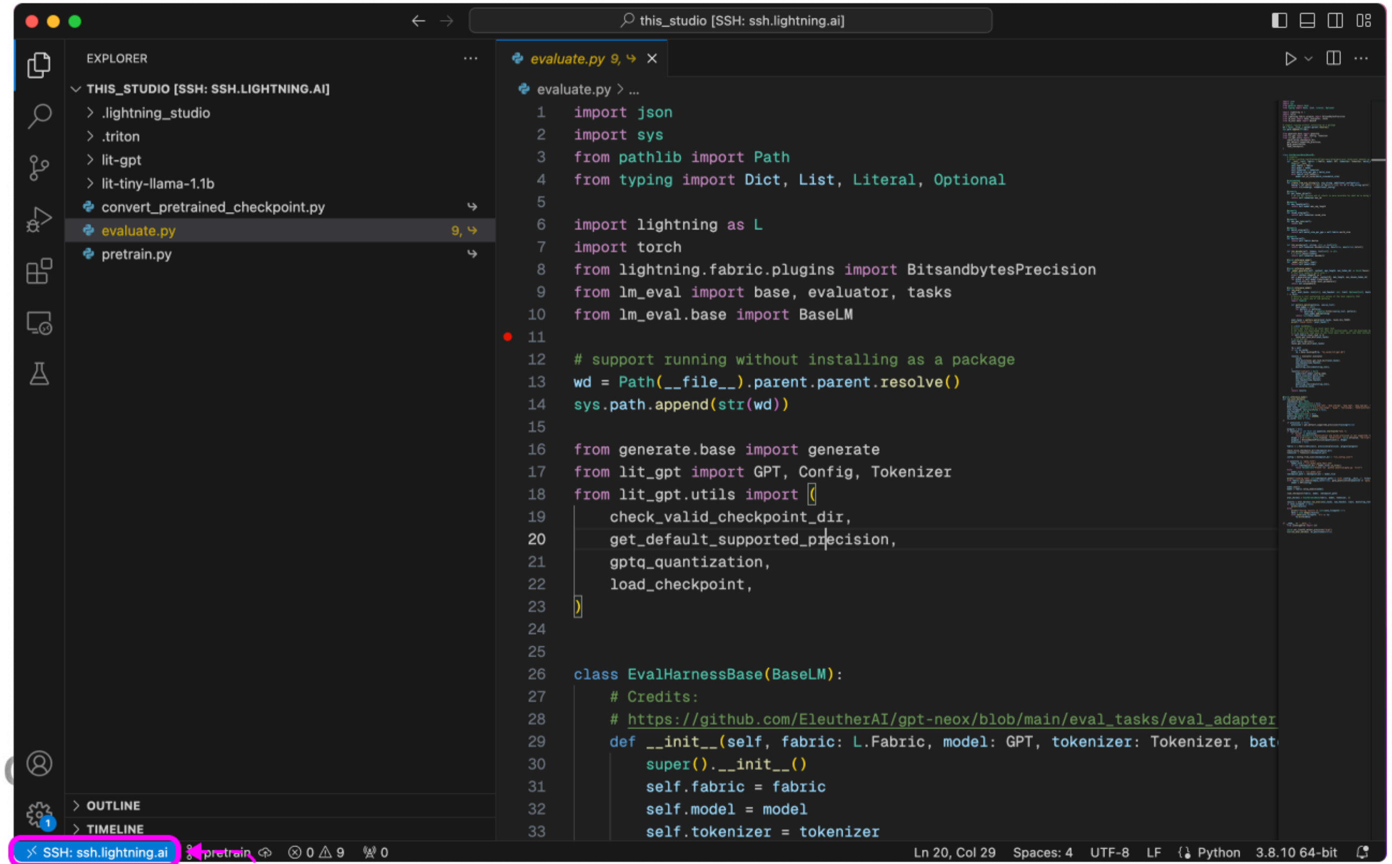
Kaggle es una plataforma en línea muy popular para competencias de **ciencia de datos y aprendizaje automático**, así como una comunidad para entusiastas y profesionales de los datos. Proporciona un espacio donde individuos y equipos pueden abordar desafíos de datos del mundo real, colaborar en proyectos y mostrar sus habilidades.



https://www.youtube.com/c/kaggle

Lightning

Lightning es un marco de trabajo (framework) de alto nivel para el desarrollo de modelos de aprendizaje profundo en Python, diseñado para simplificar y estandarizar el proceso de entrenamiento de redes neuronales. Construido sobre PyTorch, Lightning abstrae la lógica de entrenamiento, validación y prueba, permitiendo al desarrollador concentrarse en el diseño del modelo y el análisis de resultados. Se caracteriza por su estructura clara, modularidad y escalabilidad, facilitando la escritura de código limpio, reproducible y eficiente, así como la ejecución de experimentos en diferentes dispositivos y arquitecturas de cómputo.

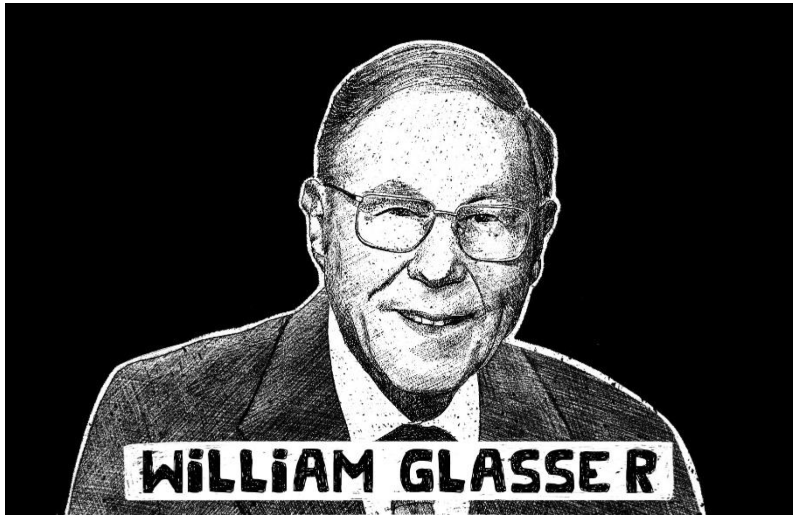


IMPORTANTE

```
Beautiful is better than ugly.
Explicit is better than implicit.
Simple is better than complex.
Complex is better than complicated.
Flat is better than nested.
Sparse is better than dense.
Readability counts.
```

Listing 2.1: Sample of the Zen of Python.

Introducción



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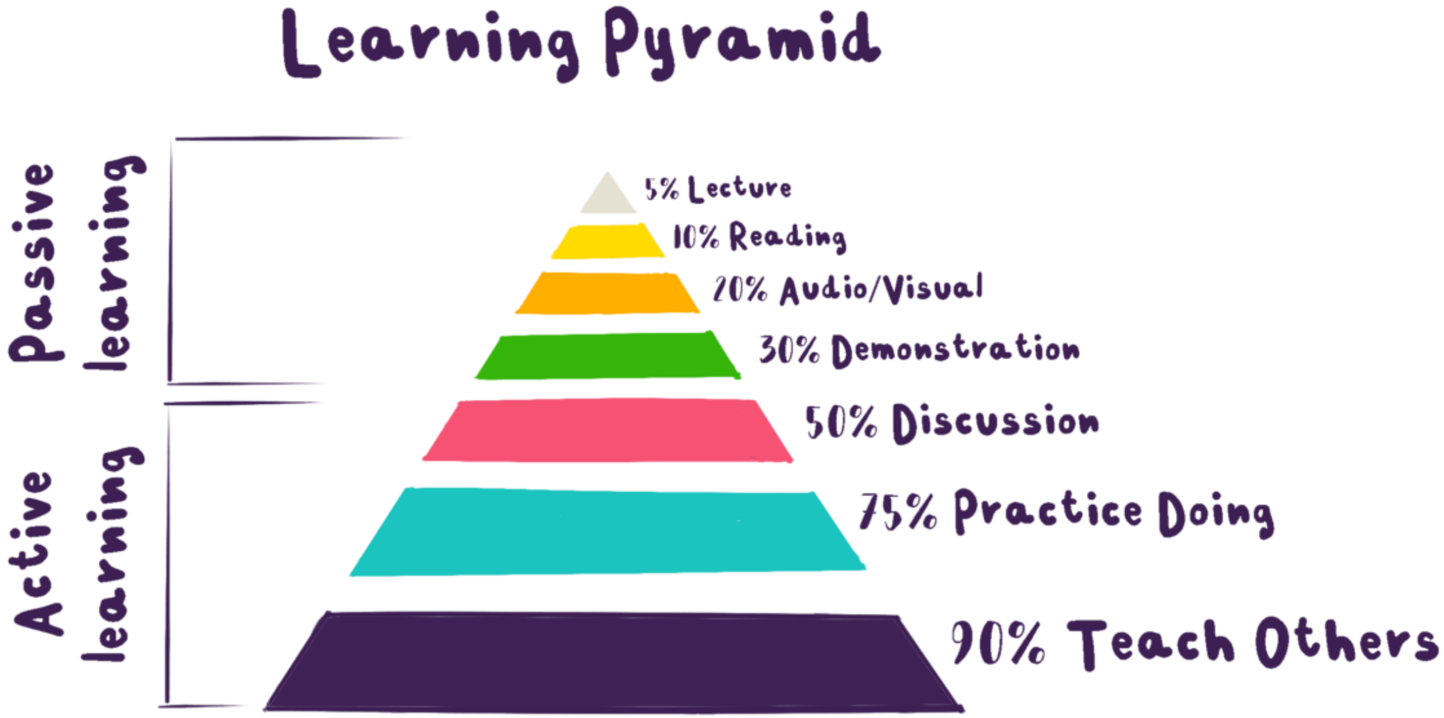
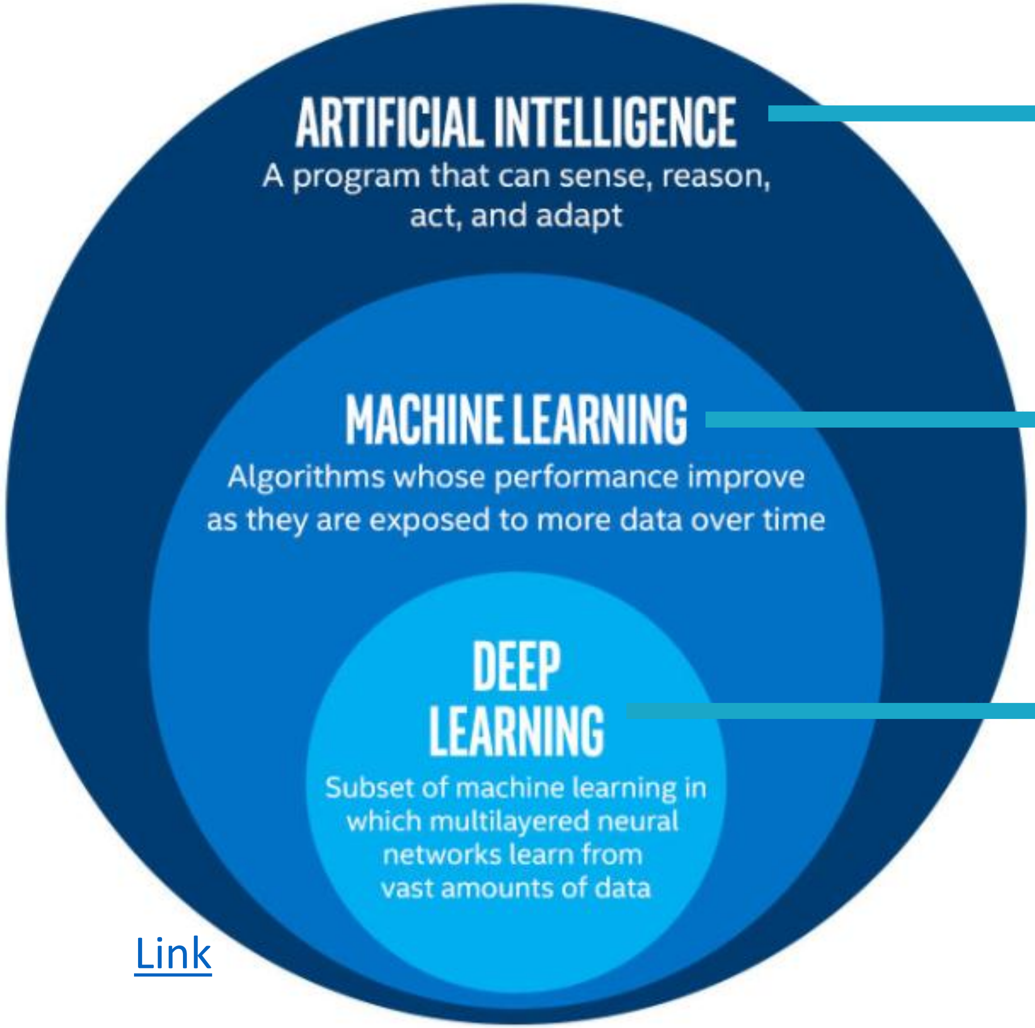


Link

Pequeñas o grandes diferencias!!!



Link

A technique which enables machine to mimic human behavior

Subset of AI technique which use statical models to enable machines to improve with experience

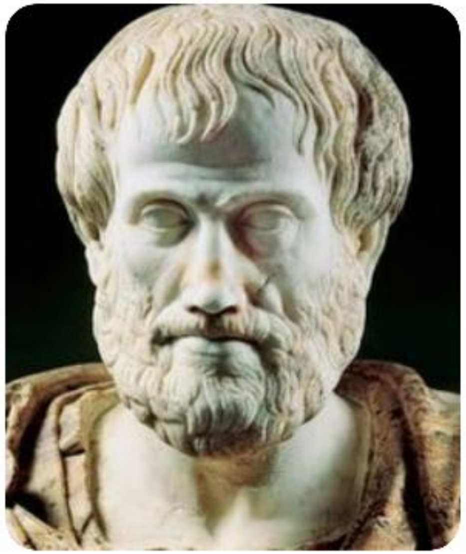
Subset of ML that uses neural networks to evaluate various factors with a similar framework to a human neural system. It has networks that can learn from unstructured or unlabeled data without supervision.

¿Qué es la inteligencia?

La inteligencia es un concepto complejo que puede definirse como la **capacidad de aprender, comprender, razonar, resolver problemas y adaptarse a nuevas situaciones**.

Implica la habilidad de **percibir información, retenerla como conocimiento y aplicarla de manera efectiva**.

Does intelligence imply knowledge?

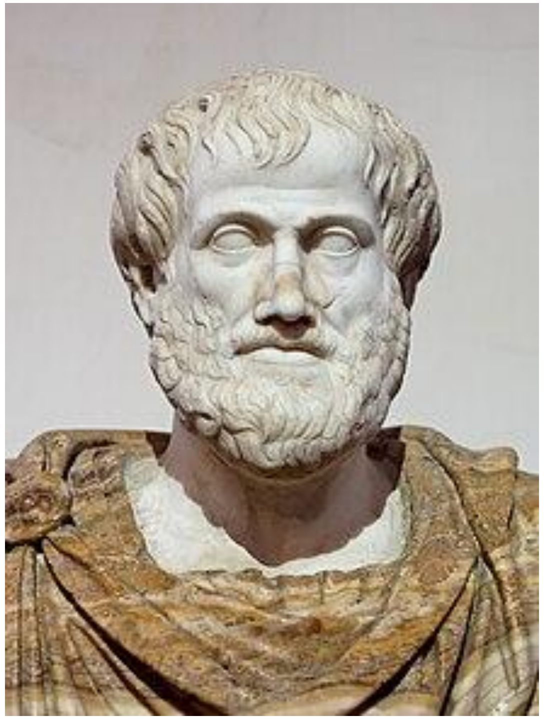


Aristotle

Understanding – awareness- be

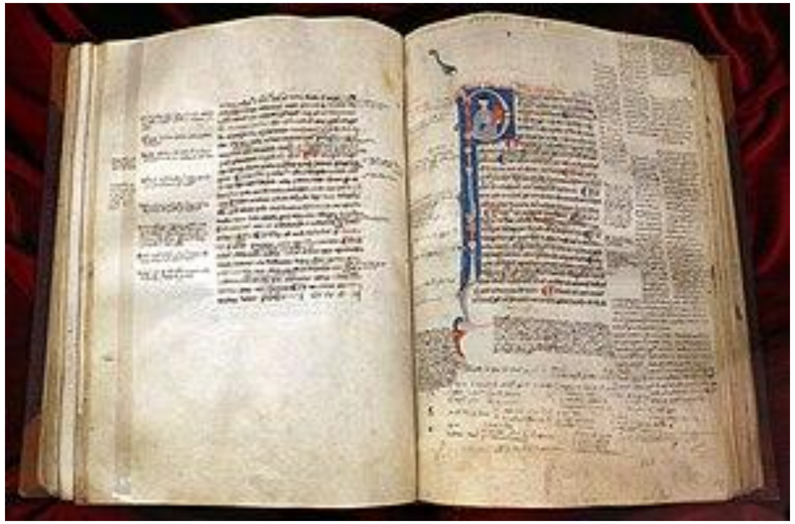
The intelligence consists not only in the knowledge but also in the skill to apply the knowledge into practice.

Breve Historia y contexto



Aristotle 380 BC
Prior Analytics / syllogistic

Un silogismo es un tipo de razonamiento lógico que consta de dos proposiciones llamadas premisas y una conclusión.
Por ejemplo:
Premisa 1: Todos los seres humanos son mortales.
Premisa 2: Sócrates es humano.
Conclusión: Por lo tanto, Sócrates es mortal.



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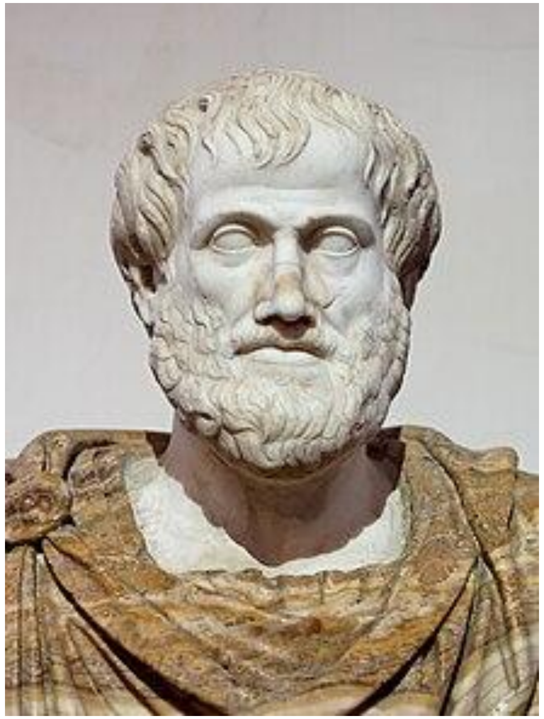

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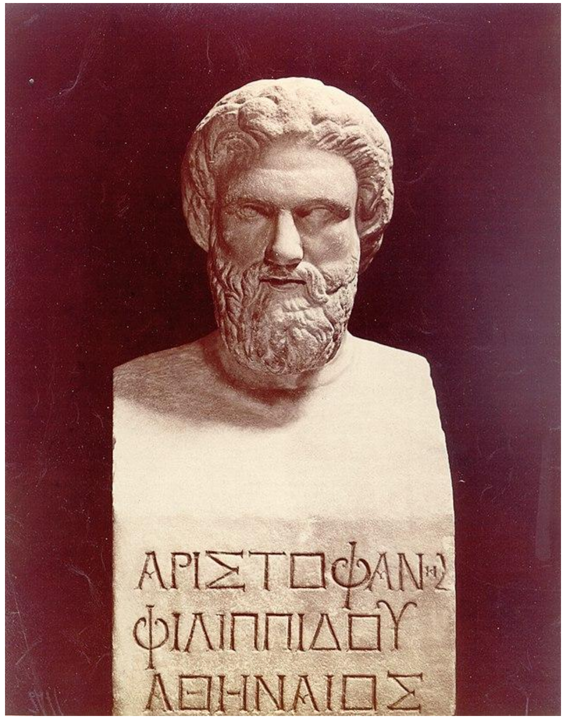
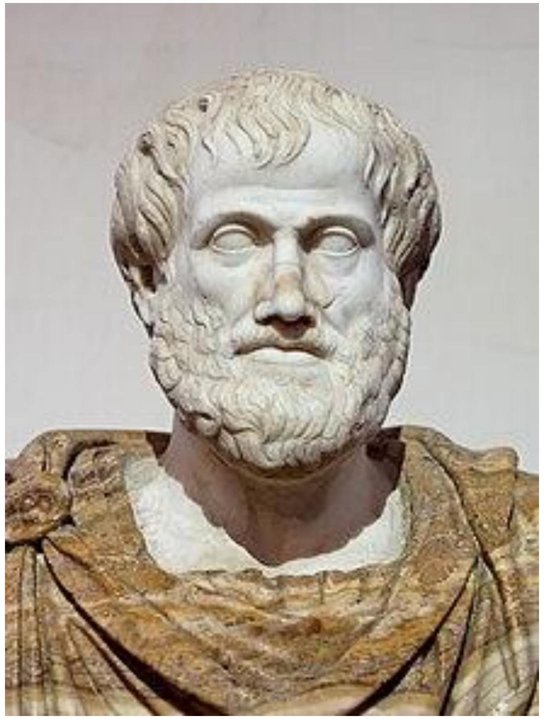


Aristophanes 300 BC
Machines that performed tasks performed by humans

Breve Historia y contexto



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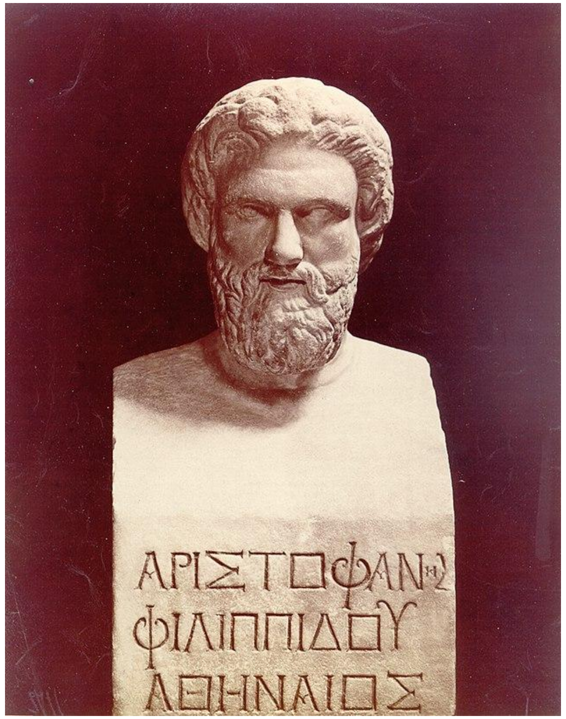

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Ctesibio 200 BC
Horologium ex aqua / clepsidra

Breve Historia y contexto

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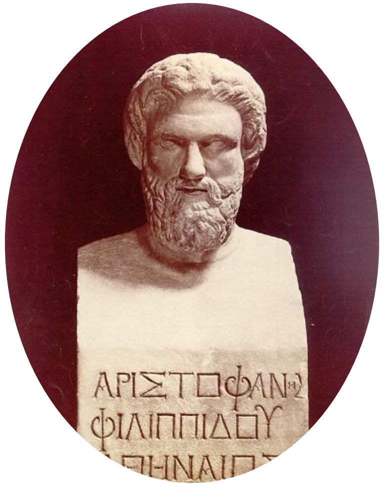

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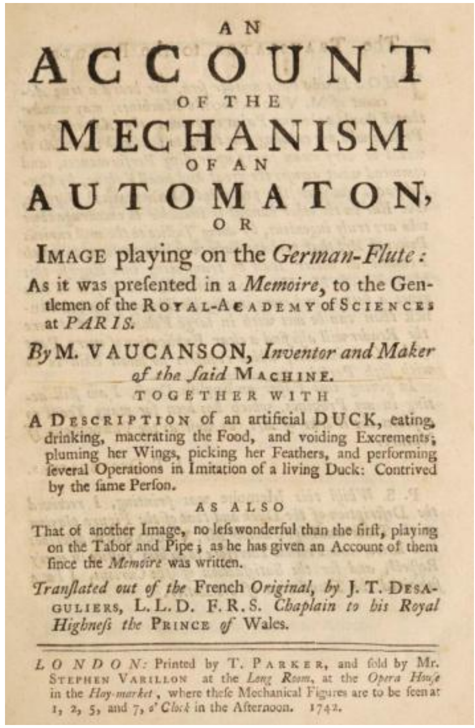


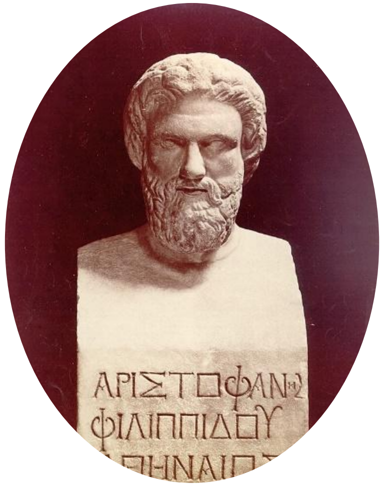
AN
ACCOUNT
OF THE
MECHANISM
OF AN
AUTOMATON,
OR
IMAGE playing on the *German-Flute*:
As it was prefented in a *Memoire*, to the Gentlemen of the ROYAL-ACADEMY of SCIENCES at *PARIS*.
By M. VAUCANSON, *Inventor and Maker of the faid* MACHINE.
TOGETHER WITH
A DESCRIPTION of an artificial DUCK, eating, drinking, macerating the Food, and voiding Excrements; pluming her Wings, picking her Feathers, and performing feveral Operations in Imitation of a living Duck: Contrived by the fame Perfon.
AS ALSO
That of another Image, no lefs wonderful than the firft, playing on the Tabor and Pipe; as he has given an Account of them fince the *Memoire* was written.
Tranflated out of the French *Original*, *by* J. T. DESAGULIERS, L.L.D. F.R.S. *Chaplain to his Royal Highnefs the* PRINCE *of* Wales.

LONDON: Printed by T. PARKER, and fold by Mr. STEPHEN VARILLON at the *Long Room*, at the *Opera House* in the *Hay-market*, where thefe Mechanical Figures are to be feen at 1, 2, 5, and 7, o'*Clock* in the Afternoon. 1742.

Jacques de Vaucanson
1764 Digesting Duck

Breve Historia y contexto

Aristotle 380 BC
Prior Analytics / syllogistic



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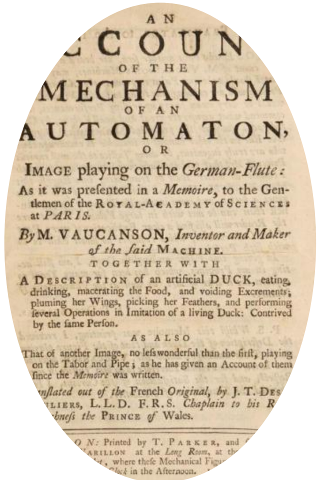

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Karel Čapek
1920 - Rossum's Universal Robots

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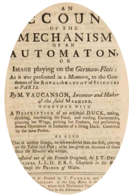

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1920 - Rossum's Universal Robots

Elektro 1939
Firth humanoid robot built by the Westinghouse Electric Corporation

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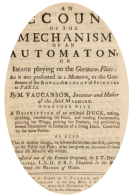

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Ada Byron

She wrote the first algorithm designed to be processed by a machine (Analytical Engine)

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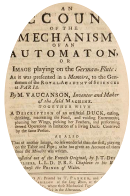

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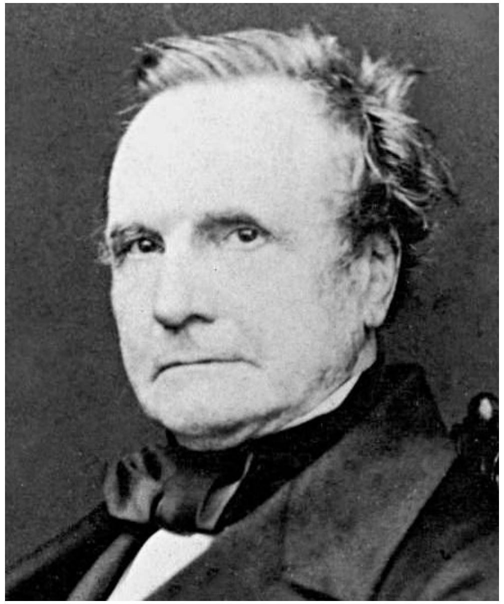

Charles Babbage

Pioneer of the Analytical Engine
1822 The beautiful Fragment

https://computerhistory.org/blog/the-analytical-engine-28-plans-and-counting/
https://www.cs.virginia.edu/~robins/Ada_and_the_First_Computer.pdf

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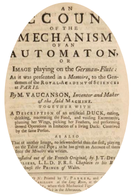

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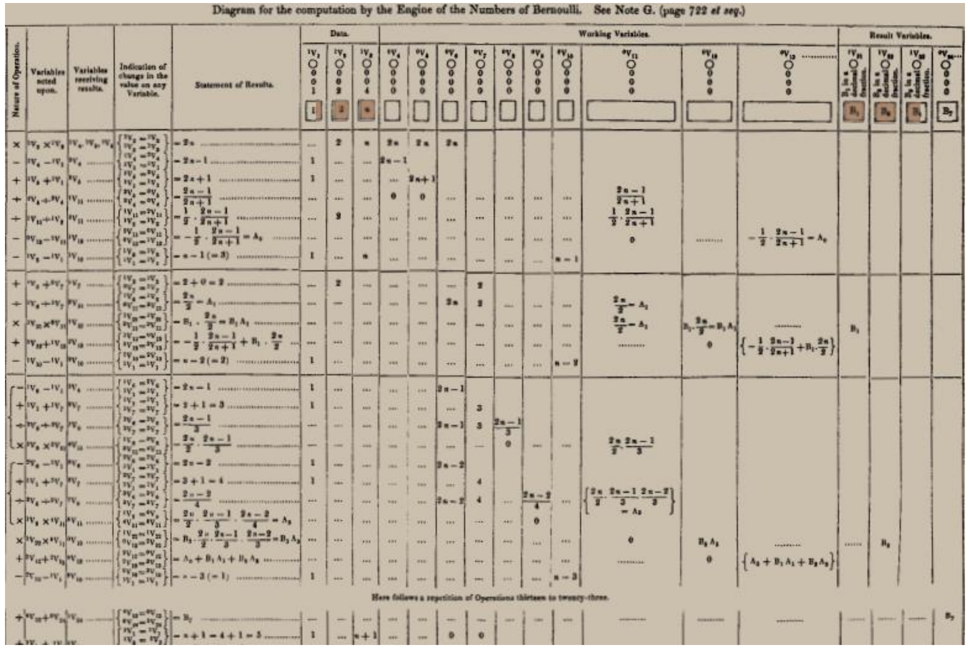

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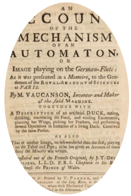

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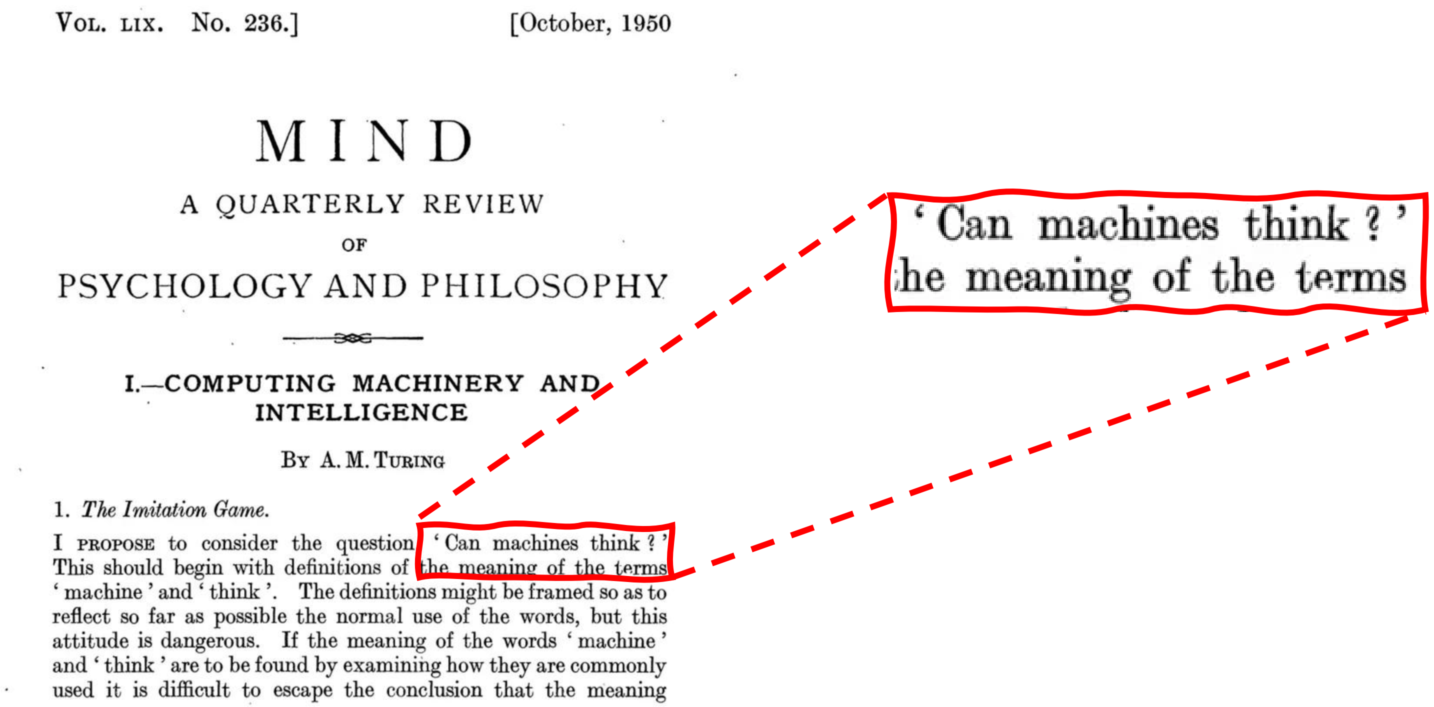

Vol. LIX. No. 236.] [October, 1950

M I N D

A QUARTERLY REVIEW

OF

PSYCHOLOGY AND PHILOSOPHY

I.—COMPUTING MACHINERY AND INTELLIGENCE

By A. M. Turing

1. *The Imitation Game.*

I propose to consider the question, 'Can machines think?' This should begin with definitions of the meaning of the terms 'machine' and 'think'. The definitions might be framed so as to reflect so far as possible the normal use of the words, but this attitude is dangerous. If the meaning of the words 'machine' and 'think' are to be found by examining how they are commonly used it is difficult to escape the conclusion that the meaning

'Can machines think?'
the meaning of the terms

The imitation game or Turing Test is a method of inquiry in artificial intelligence (AI) for determining whether or not a computer is capable of thinking like a human being

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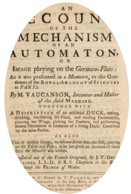

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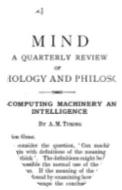
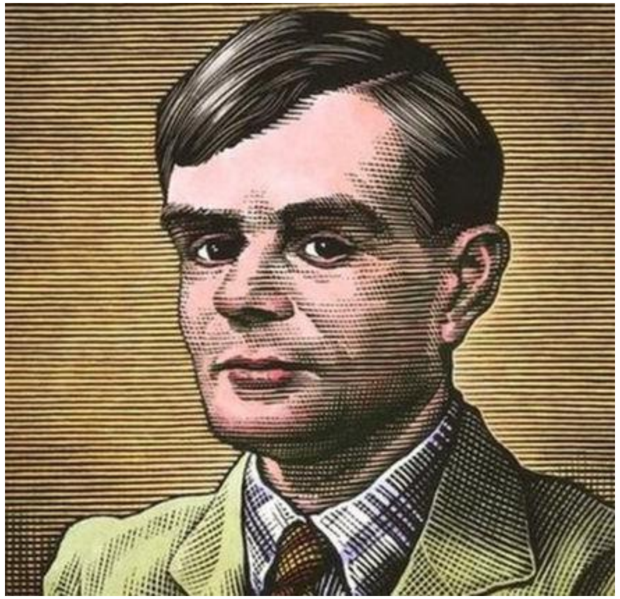
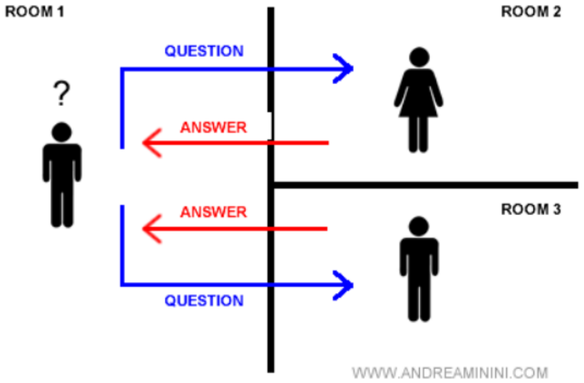
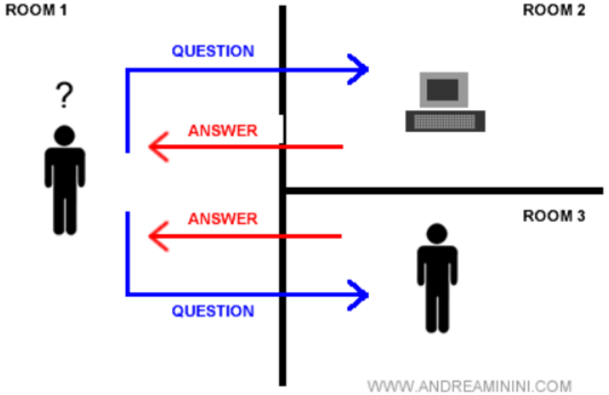


Turing Test 1950
Can Machines Think_

The imitation game or Turing Test is a method of inquiry in artificial intelligence (AI) for determining whether or not a computer is capable of thinking like a human being







Breve Historia y contexto

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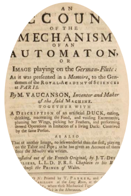

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1920 - Rossum's Universal Robots

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Ada Byron
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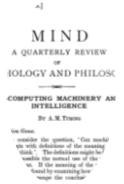

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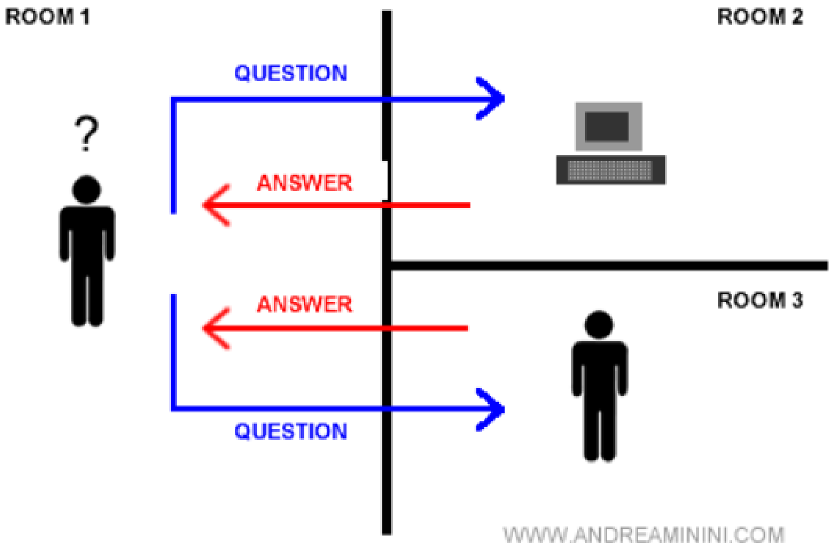
Can Machines Think_



ELIZA—a computer program for the study of natural language communication between man and machine

J. Weizenbaum • Published in CACM 1 January 1966 • Computer Science • Communications of the ACM

Joseph Weizenbaum

ELIZA — A Computer Program for the Study of Natural Language Communication between Man and Machine in 1966.

https://datatracker.ietf.org/doc/html/rfc439

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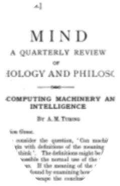

Turing Test 1950
Can Machines Think_

Enigma machine

https://www.youtube.com/watch?v=zCn3GCOwmeI

An electromechanical encryption device primarily used by the armed forces of Nazi Germany during World War II.

The **Turing machine** is a theoretical mathematical model of computation capable of performing any computation that can be carried out by any modern digital computer.

ON COMPUTABLE NUMBERS, WITH AN APPLICATION TO THE ENTSCHEIDUNGSPROBLEM

By A. M. TURING.

[Received 28 May, 1936.—Read 12 November, 1936.]

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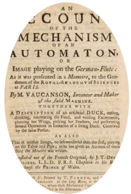

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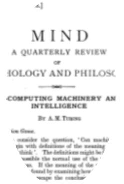

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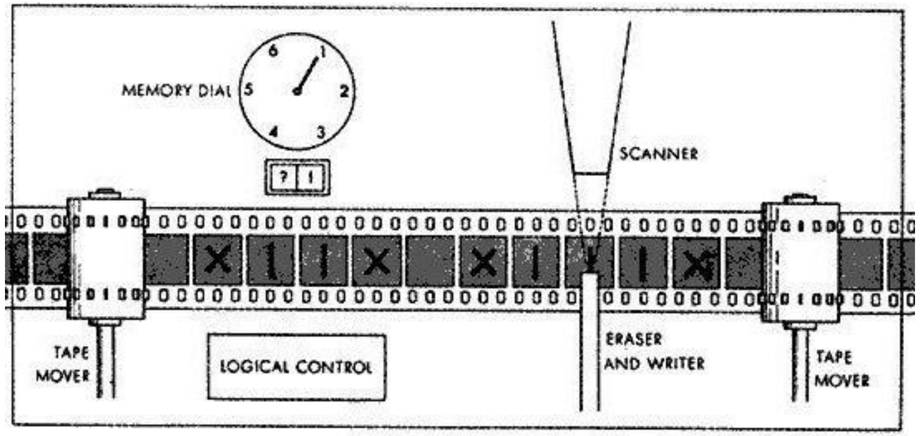
Is there a universal algorithm that, given any mathematical statement, determines in a finite number of steps whether it is true or false?



Breve Historia y contexto

Kurt Gödel

Incompleteness theorem "Any effectively generated formal system capable of expressing elementary arithmetic cannot be both consistent and complete. In particular, for any such consistent formal system, there will exist statements that are undecidable within the system." 1931

https://www.youtube.com/shorts/A-GV6LduJNM

David Hilbert

"**Decision problem**" is related to the question of whether there is an algorithm that can determine whether a given mathematical statement is true or false, 1928.

Entscheidungsproblem

The halting problem is a fundamental question in the theory of computing. The problem refers to the impossibility of determining, by one algorithm, whether another algorithm will stop its execution or enter an infinite loop for any given input. 1936

Halting problem

Breve Historia y contexto

The halting problem is a fundamental question in the theory of computing. The problem refers to the impossibility of determining, by one algorithm, whether another algorithm will stop its execution or enter an infinite loop for any given input. 1936

Halting problem

WALK

```
Input: x

steps = 0
while steps != x:
  move forward
  turn right
  steps = steps + 1
```

HALT

```
Inputs: program, input

if program halts
when given input:
  return yes
else:
  return no
```

Proof by Contadiction

1. Assume there is a program that solves the halting problem.
2. Show that the assumption leads to a contradiction, so the assumption must not be true.

https://www.youtube.com/watch?v=Kzx88YBF7dY

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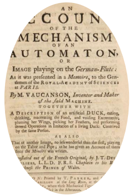

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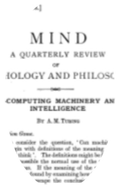

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ELIZA — A Computer Program for the Study of Natural Language Communication

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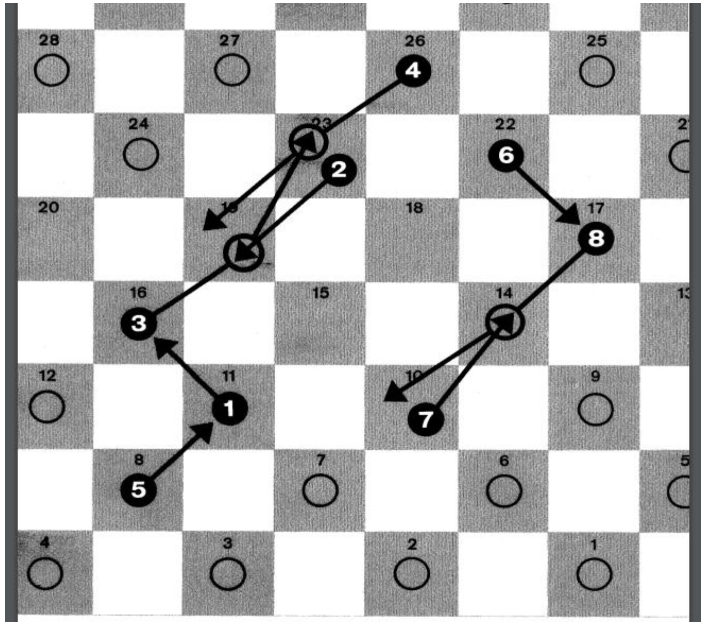
1948 / 1959 "Some Studies of Machine Learning Using the Game of Checkers"

The most interesting thing about this program is that it was able to improve its level of play as it played, adjusting the weights of the important variables for decision making.



https://ieeexplore.ieee.org/document/5392560?denied=

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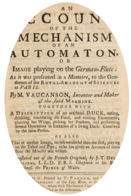

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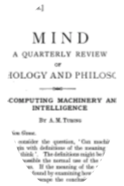

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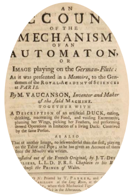

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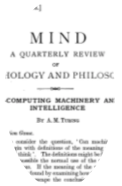

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Machine Learning is a subset of AI that involves the development of algorithms and statistical models that enable computers to learn and make predictions or decisions from data without being explicitly programmed..

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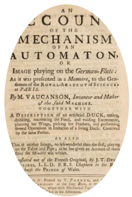

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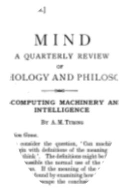

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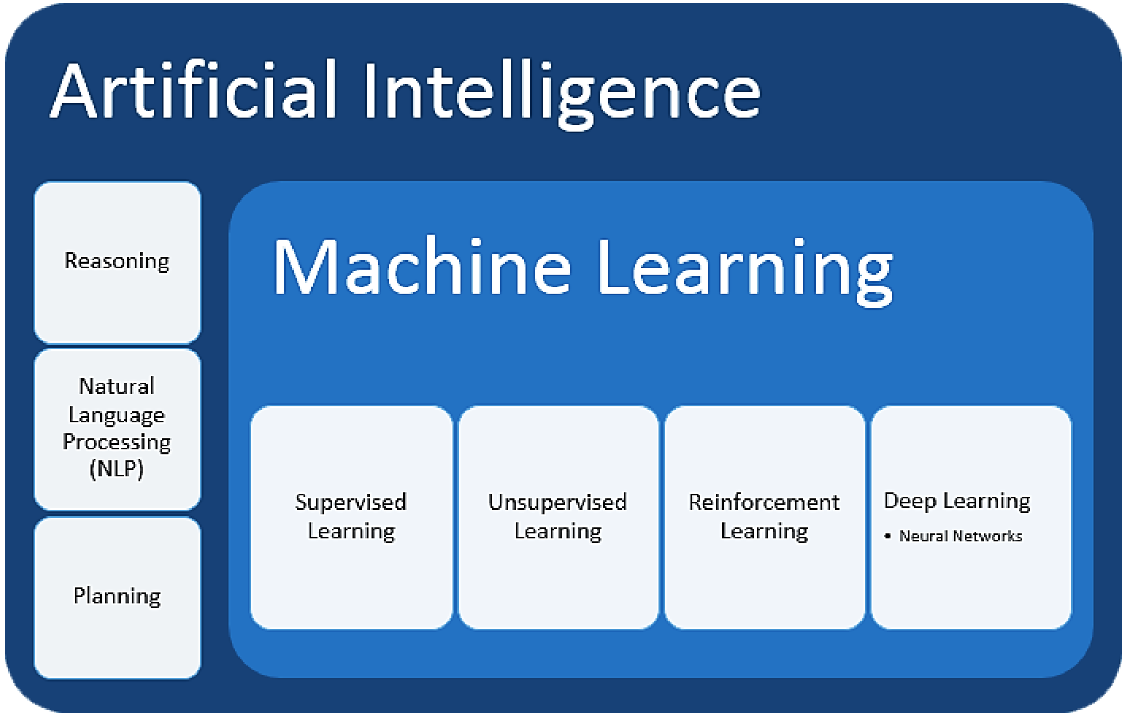
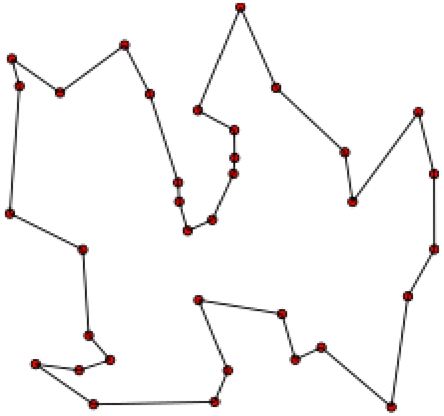


www.ibm.com/co-es/analytics/machine-learning



Nearest Neighbor

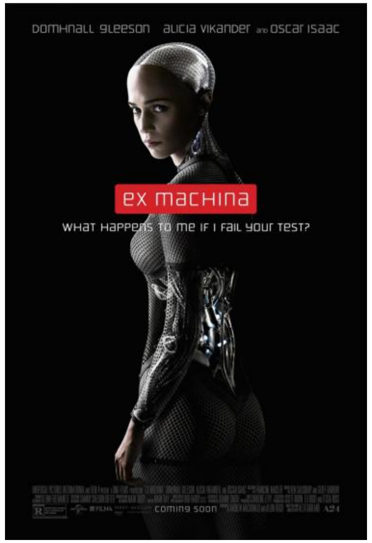
1967 It is considered the first pattern recognition algorithm

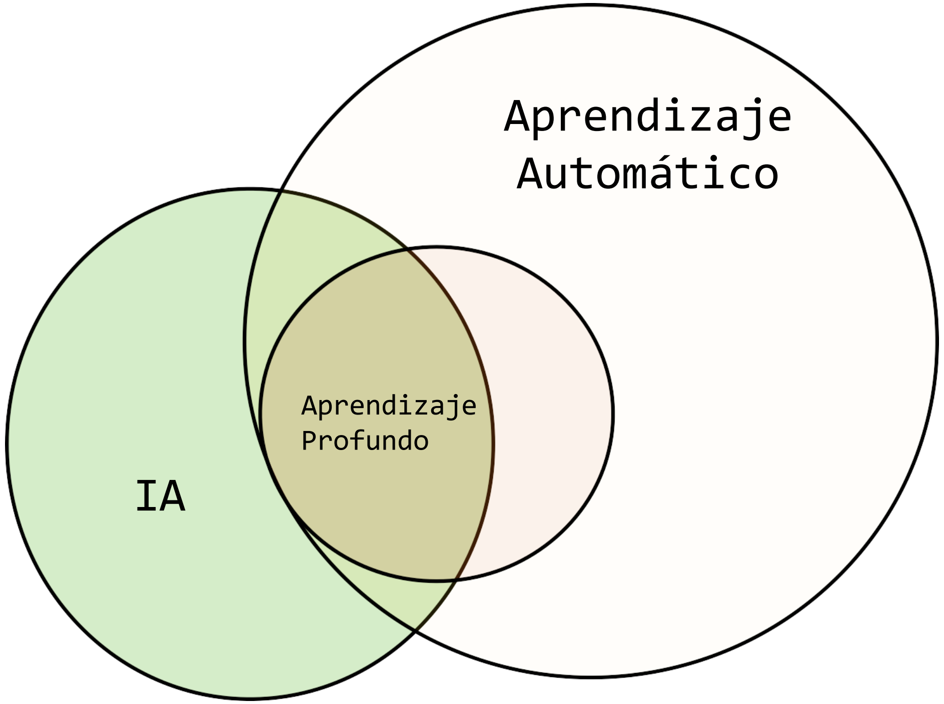
Breve Historia y contexto

Inteligencia Artificial (IA): Subcampo de la informática centrado en resolver tareas que los humanos realizan con facilidad.

IA Estrecha (narrow): Diseñada para resolver tareas específicas de forma eficiente.

Inteligencia Artificial General (AGI): IA capaz de abordar múltiples tareas y comportarse como un humano en diversos contextos.





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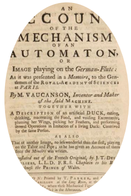

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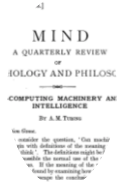

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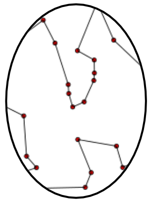

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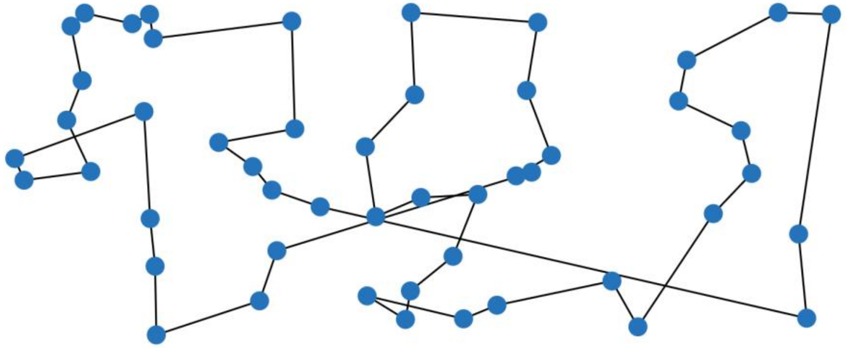


The traveling salesman problem (TSP) asks the following question: "Given a list of cities and the distances between each pair of cities, what is the shortest possible route that visits each city exactly once and returns to the origin city?

https://www.math.uwaterloo.ca/tsp/

Inteligencia Artificial

Input

"Hey Google play some music"

Google assistant

Processing data

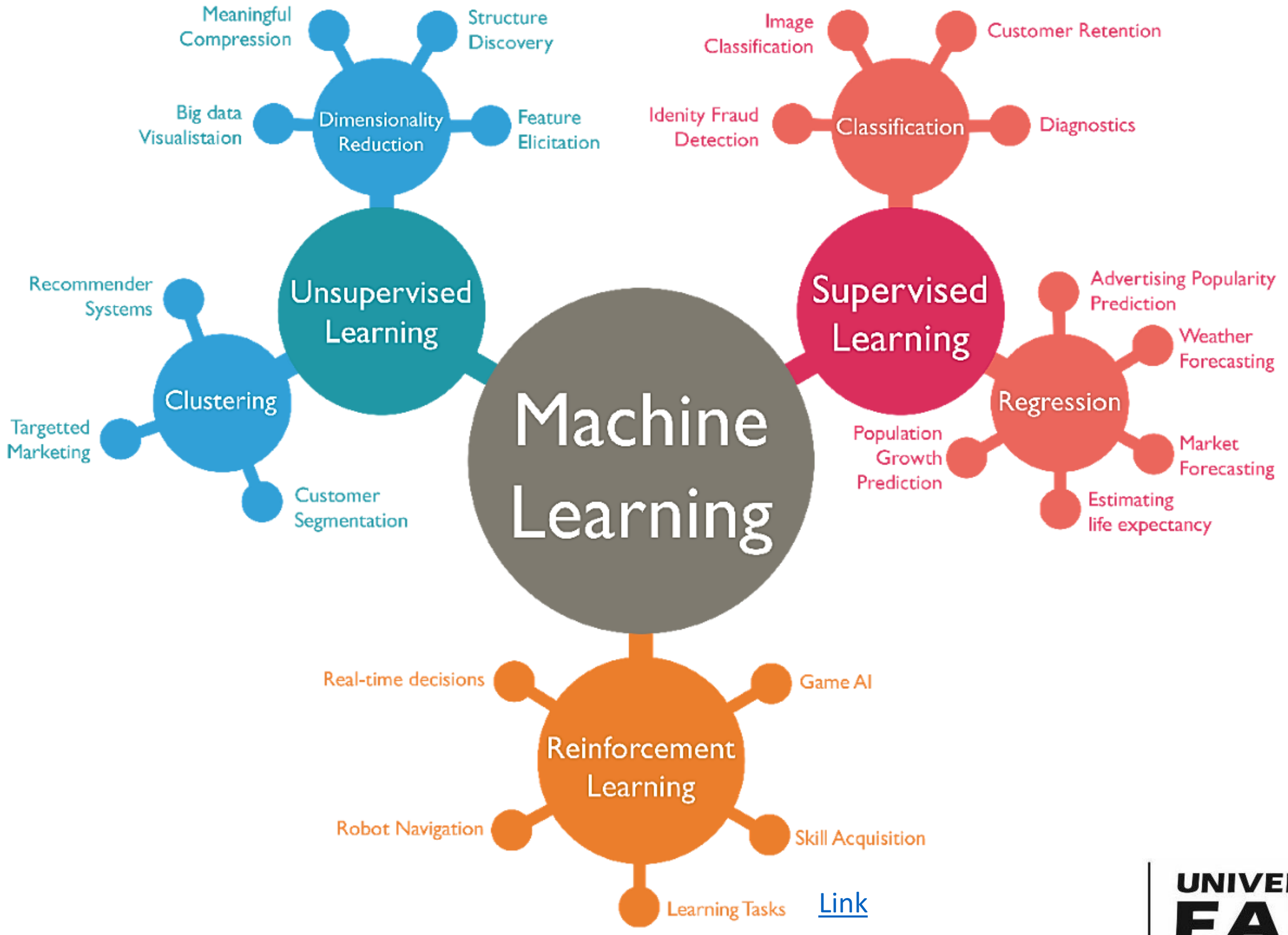
```
<!--shortcuts.xml-->
  <capability
      <intent
          ...
      </intent>
  </capability>
```

Output

Qué es Machine Learning?

Machine Learning Es un subconjunto de la Inteligencia Artificial que implica el desarrollo de algoritmos y modelos estadísticos que permiten a las computadoras aprender y hacer predicciones o tomar decisiones a partir de datos, sin ser programadas explícitamente.

MATHEMATICAL REPRESENTATION OF A REAL DATA SYSTEM OBTAINED USING STATISTICS.



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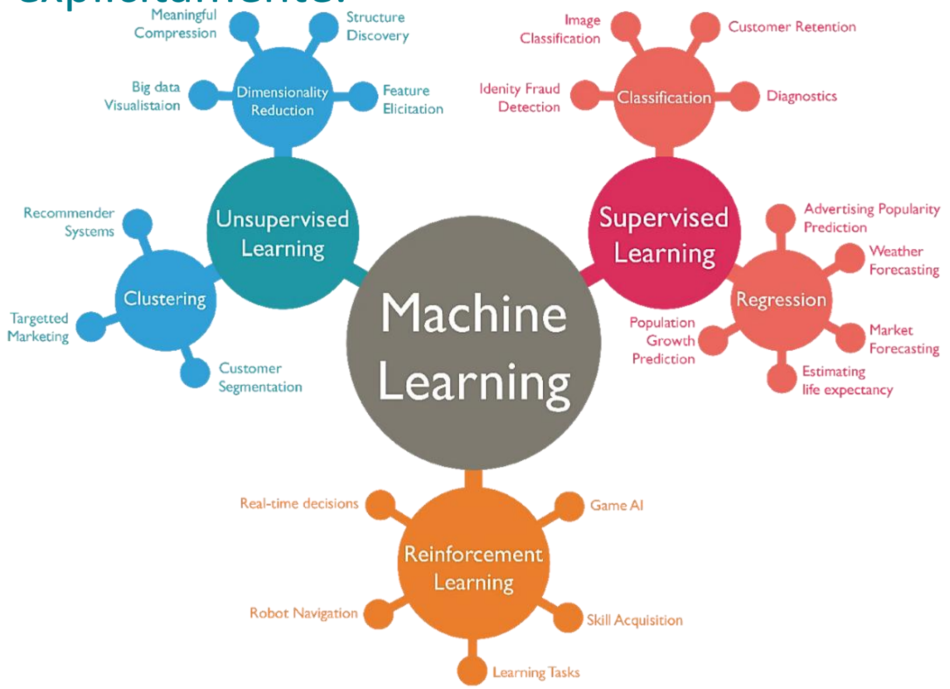
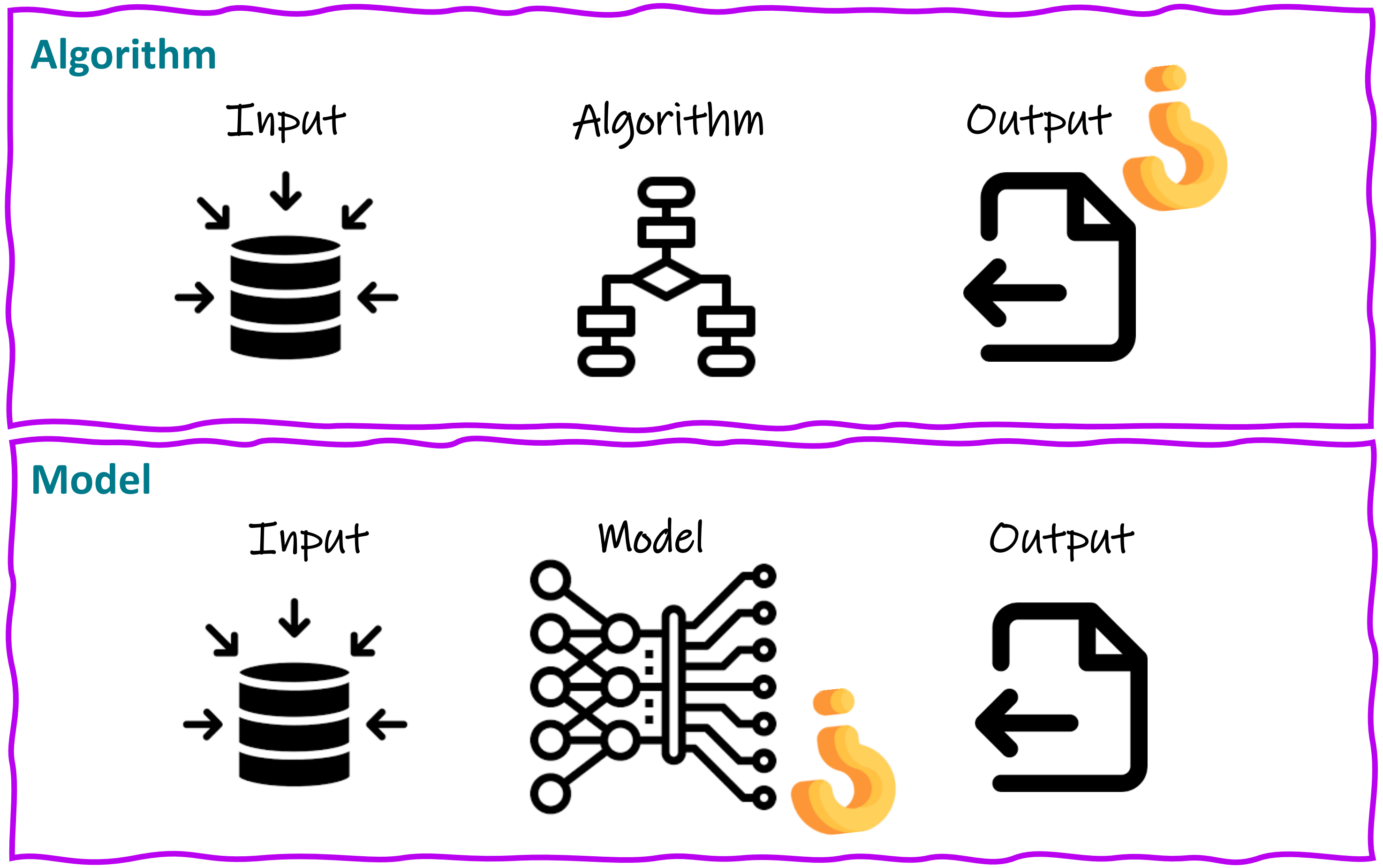




[Link]

Qué es Machine Learning?

Traditional Programming:

1. **Algorithm Development**

Algorithm Design

2. **Algorithm Implementation**

Algorithm + CODING → Program

3. **Operation**

Input Args → Program → Results

Machine Learning:

1. **Data Recollection and Processing**

Structured & Non-Structured Data → Raw Data → Processed Data

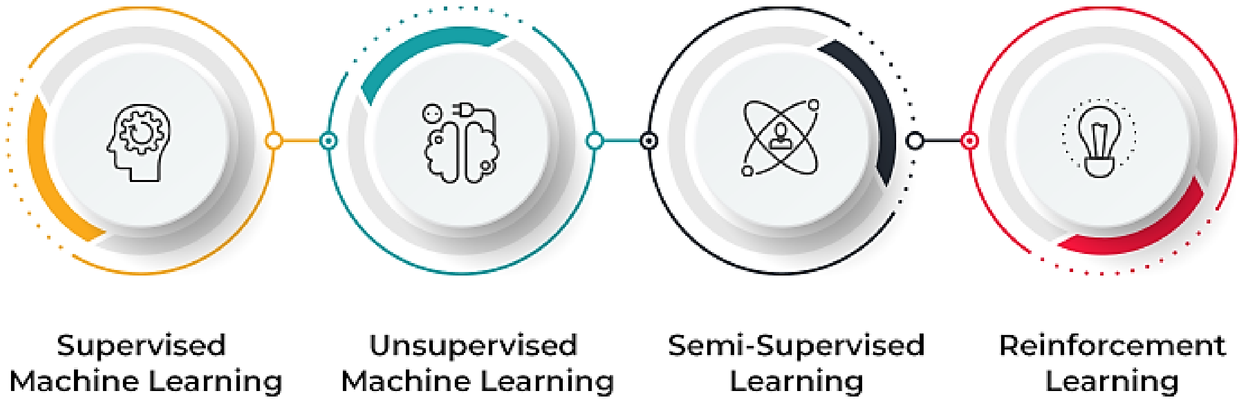
2. **Model Experimentation**

Processed Data + ML Algorithms → ML Model

3. **Operation**

Input Data → ML Model → Results

Qué es Machine Learning?



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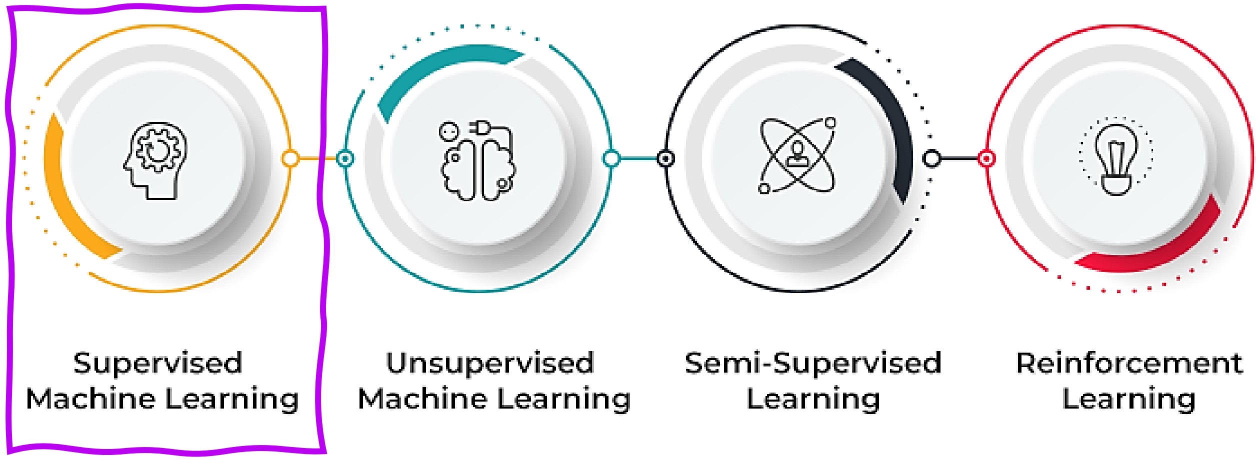
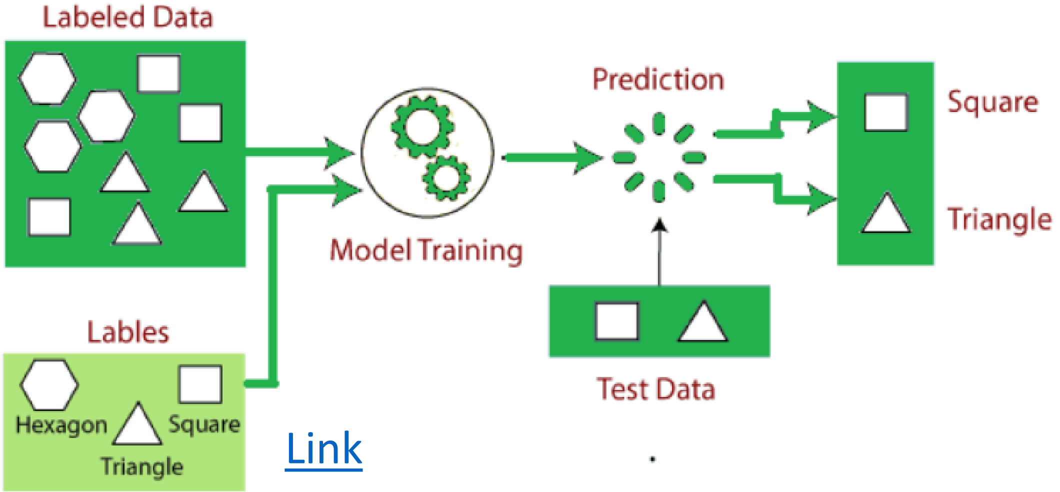
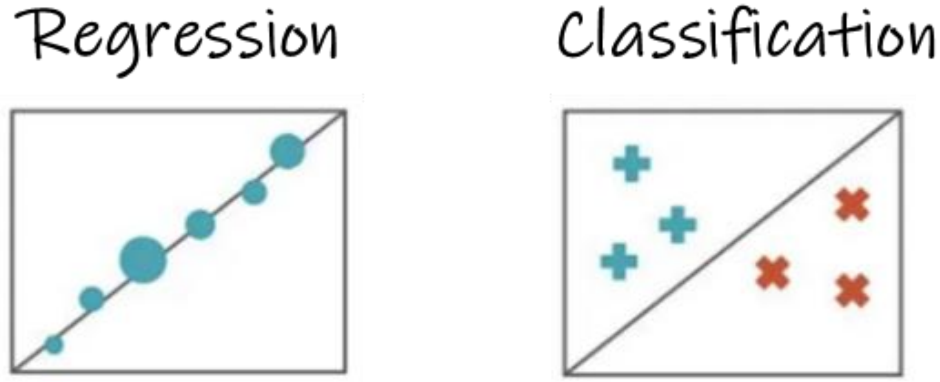




Link

Se define por el uso de conjuntos de datos etiquetados para entrenar algoritmos **capaces de clasificar datos o predecir resultados de manera precisa**.

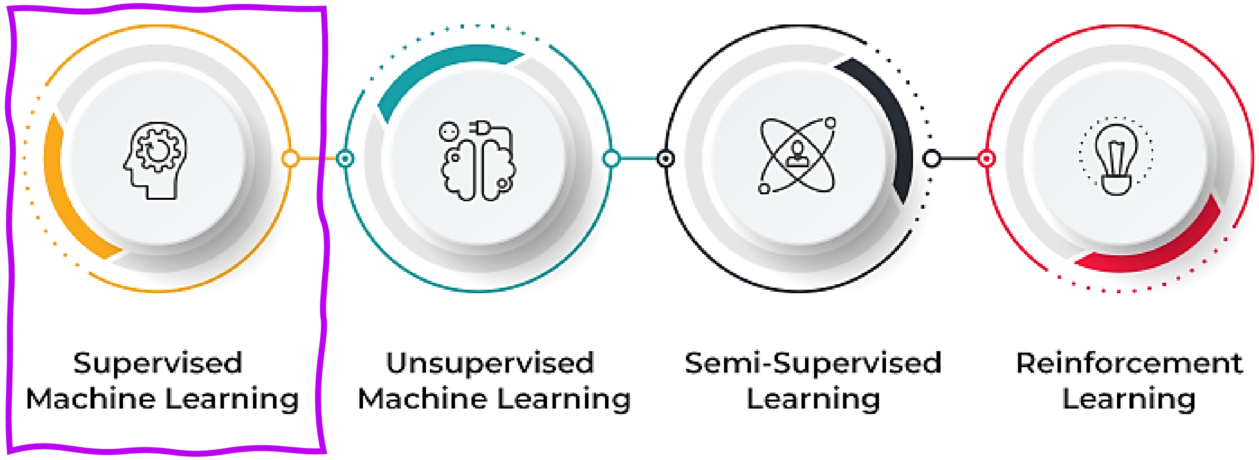
El algoritmo aprende a realizar predicciones o clasificaciones a partir de datos de entrenamiento y puede generalizar su conocimiento para hacer predicciones correctas sobre datos nuevos que no ha visto previamente.

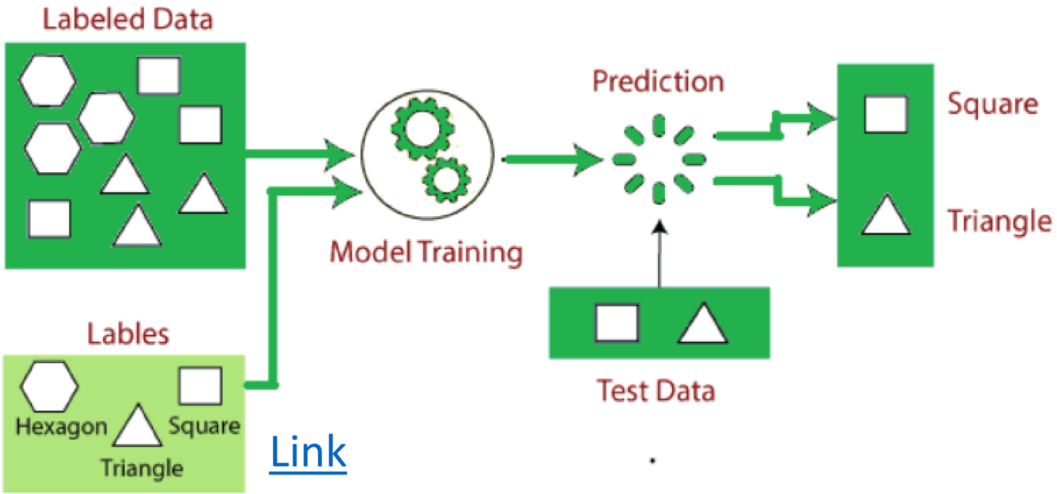


Se denomina "supervisado" porque el proceso de entrenamiento del algoritmo está **guiado** y **supervisado** por las respuestas correctas conocidas en el conjunto de datos de entrenamiento.

UNIVERSIDAD EAFIT

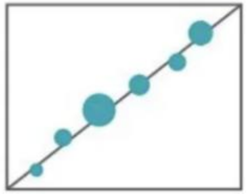
Qué es Machine Learning?



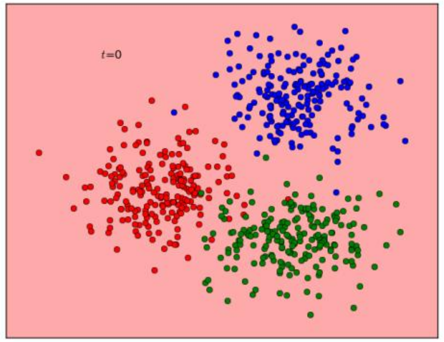


Link

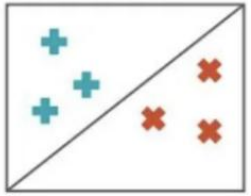
Regression



- Linear Regression
- Regression Trees
- Non-Linear Regression
- Bayesian Linear Regression
- Polynomial Regression

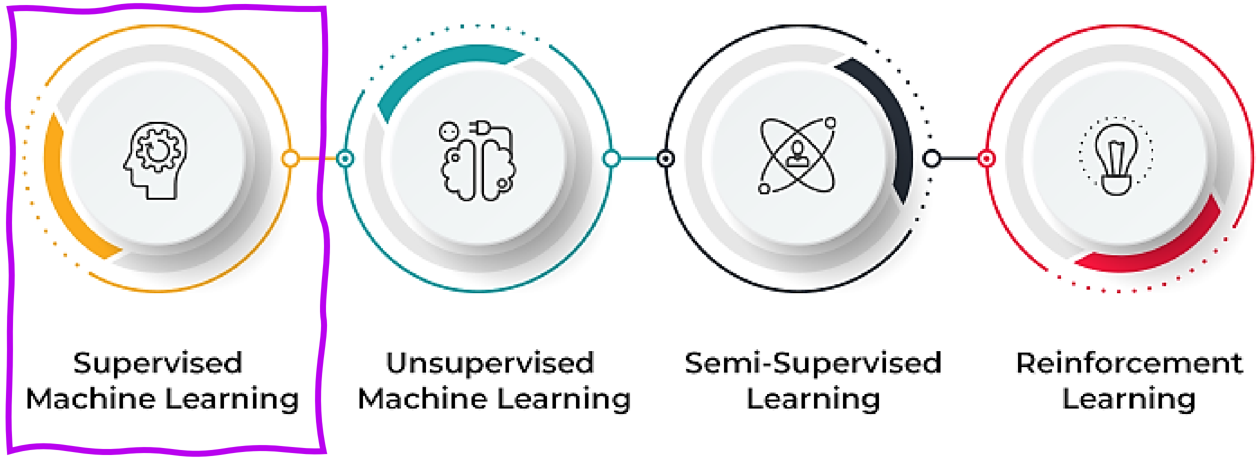


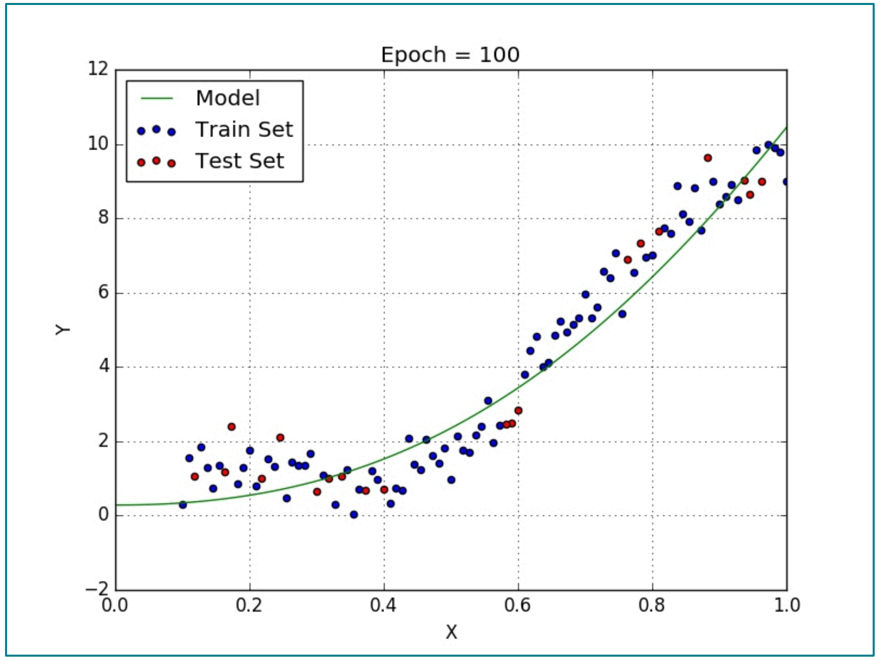
Classification



- Random Forest
- Decision Trees
- Logistic Regression
- Support vector Machines

Qué es Machine Learning?





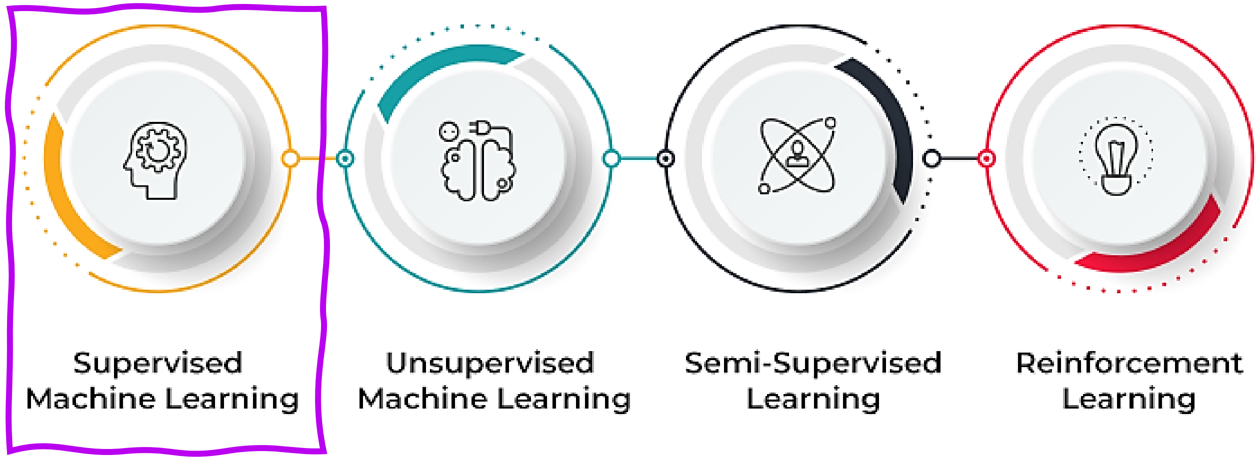
https://www.cs.toronto.edu/~frossard/post/multiple_linear_regression/

¿Cuál es la curva más eficiente que puede explicar la mayor parte de los datos?

¿Cuál es el mejor optimizador para el aproximador?

https://machinelearningmastery.com/a-tour-of-machine-learning-algorithms/

Qué es Machine Learning?

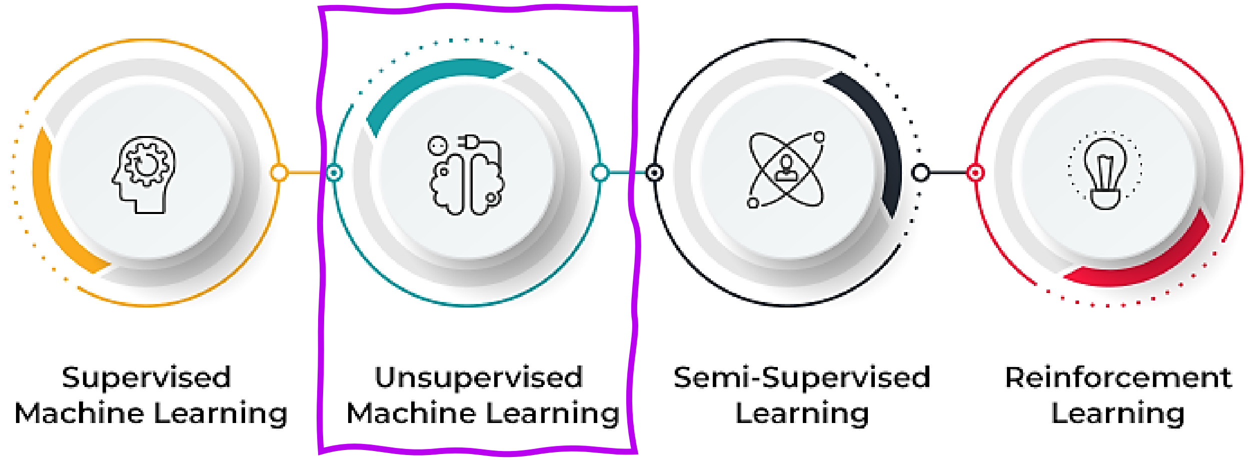


Requirements:
- No missing values
- Data in numeric format (Hot encoding)
- Data store in array

Pros:
- High Accuracy
- Easy Interpretability
- Data in numeric format (Hot encoding)
- Data store in array

Cons:
- Data Labeling
- Overfitting
- Not Scalability
- Human Intervention
- Data training period is longer, since supervised learning requires a lot of computation time.

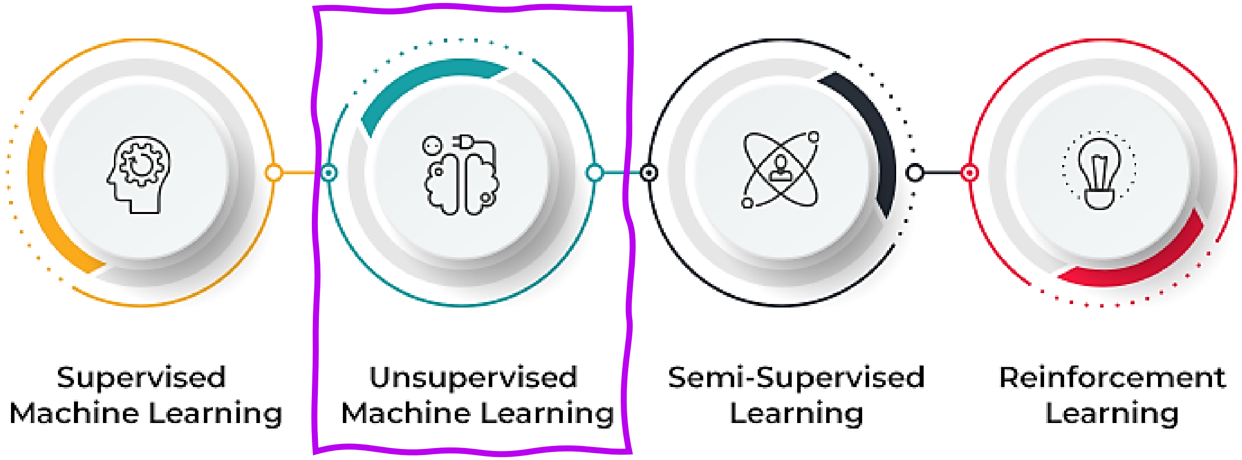
Qué es Machine Learning?



En el **aprendizaje automático no supervisado**, al algoritmo se le proporcionan datos de entrada sin ningún objetivo correspondiente o salida etiquetada. El objetivo del algoritmo es descubrir por sí mismo patrones, relaciones o estructuras dentro de los datos. Este enfoque se utiliza para tareas como la agrupación (clustering), que consiste en agrupar puntos de datos similares, y la reducción de dimensionalidad, que busca simplificar los datos preservando la información importante, sin necesidad de supervisión explícita ni de datos etiquetados.

- Clustering

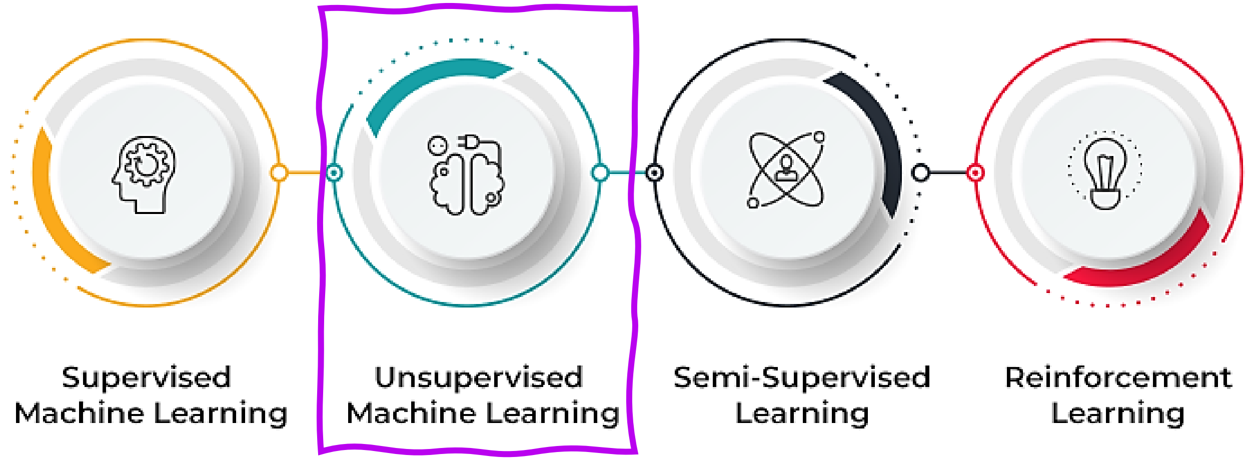
Qué es Machine Learning?

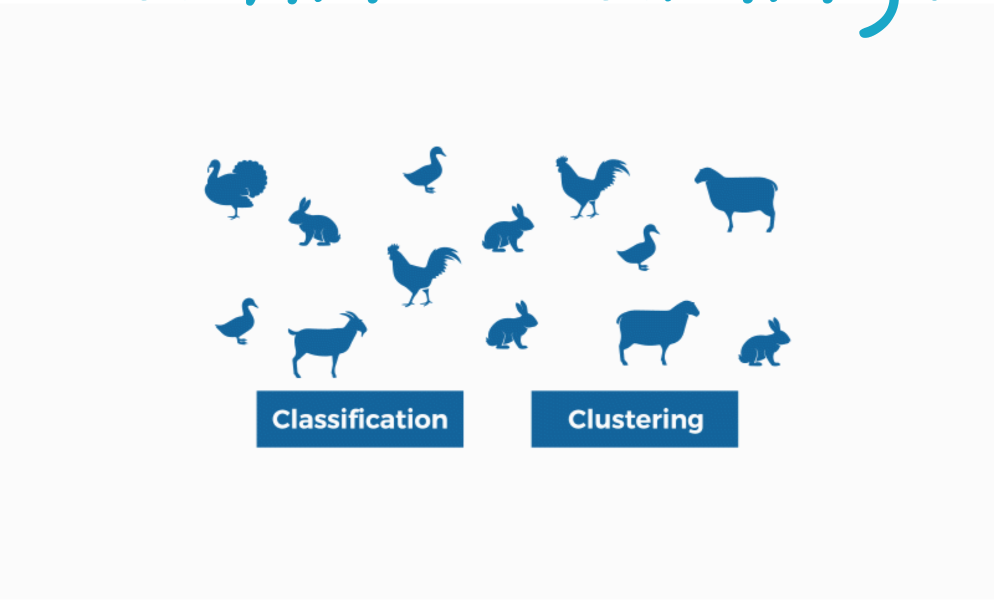


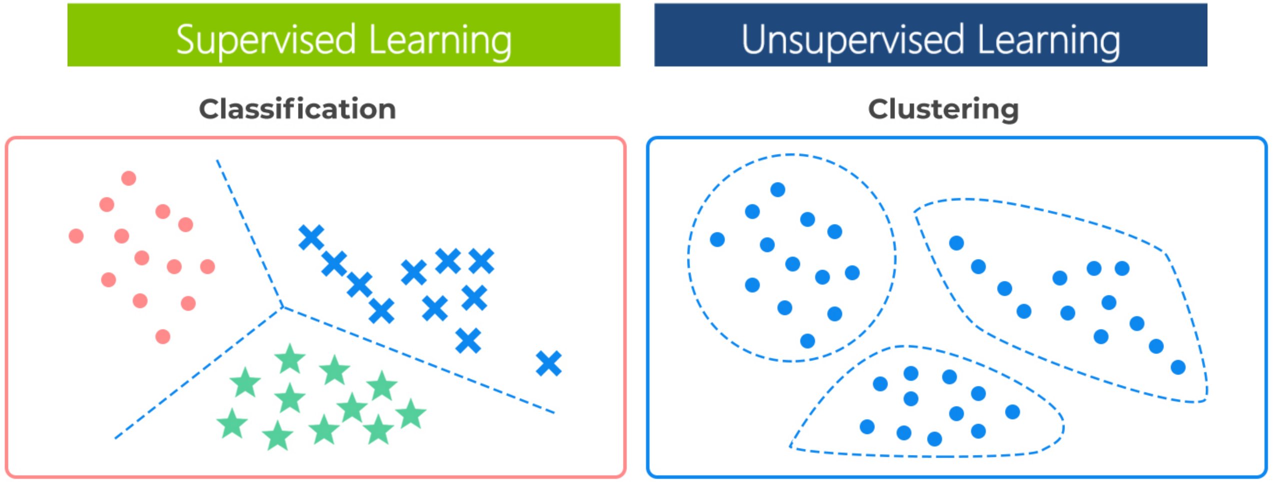
En el **aprendizaje automático no supervisado**, al algoritmo se le proporcionan datos de entrada sin ningún objetivo correspondiente o salida etiquetada. El objetivo del algoritmo es descubrir por sí mismo patrones, relaciones o estructuras dentro de los datos. Este enfoque se utiliza para tareas como la agrupación (clustering), que consiste en agrupar puntos de datos similares, y la reducción de dimensionalidad, que busca simplificar los datos preservando la información importante, sin necesidad de supervisión explícita ni de datos etiquetados.

- Clustering
- Association

Qué es Machine Learning?



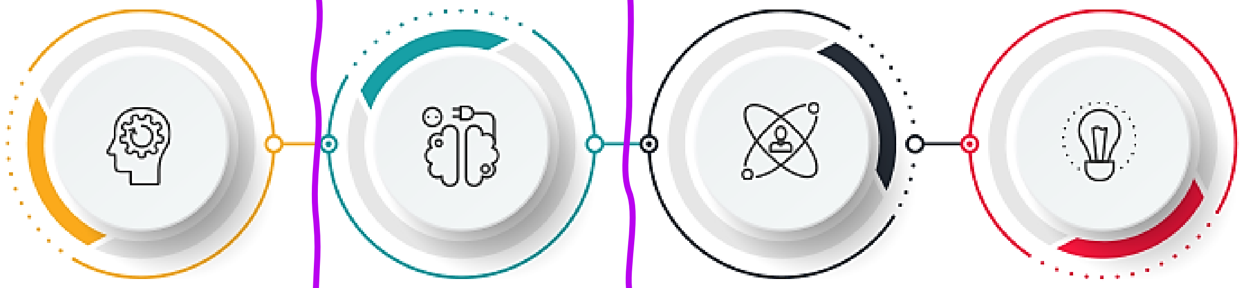




Clasificación: Predicción de etiquetas basada en características que ya han sido observadas en los datos. Permite encontrar patrones relacionados con la variable objetivo.

Clustering: Utiliza datos históricos no etiquetados para explorar y crear una estructura o forma que permita organizar dichos datos.

Qué es Machine Learning?



Supervised Machine Learning
Unsupervised Machine Learning
Semi-Supervised Learning
Reinforcement Learning

Requirements:

- No missing values
- Data in numeric format (Hot encoding)
- Data store in array

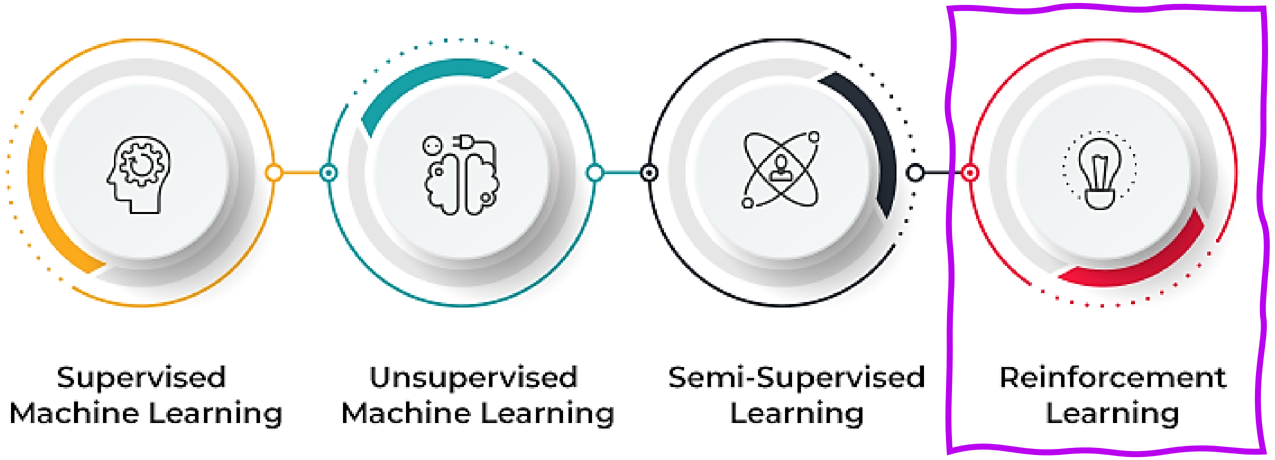
Pros:

- Learns and classifies data from no data labeling
- Requires fewer manual data preparation
- No Human Intervention
- Capable of finding previously unknown patterns in data
- Real time analysis

Cons:

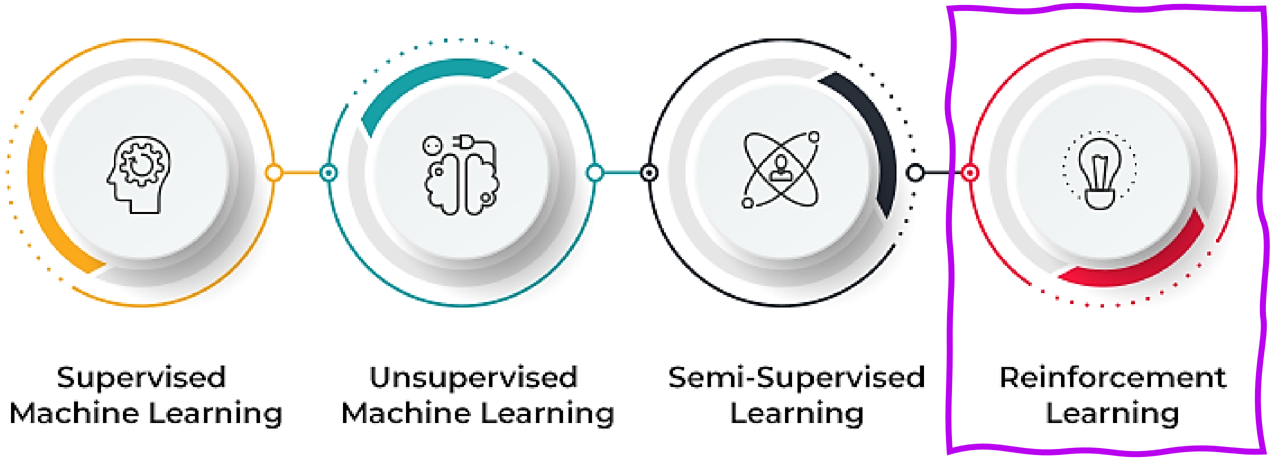
- Not Well Accuracy
- Hard to interpret (Users need to manually interpret)
- Difficult to measure accuracy or effectiveness due to lack of predefined answers during training

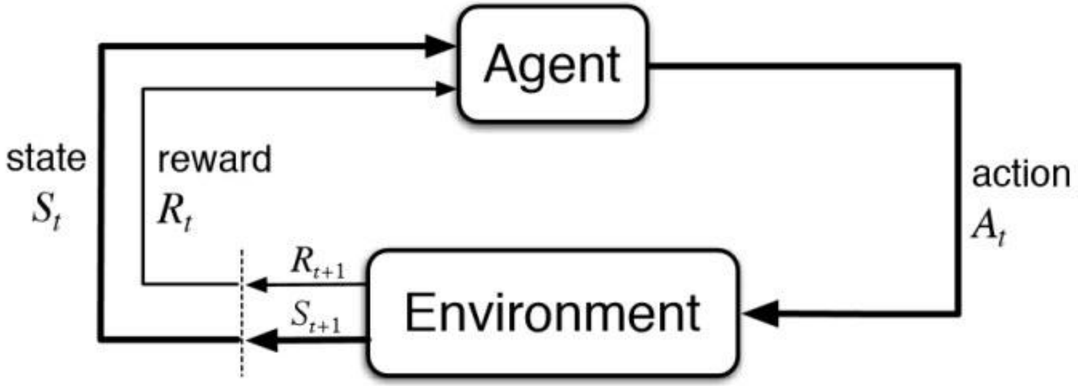
Qué es Machine Learning?

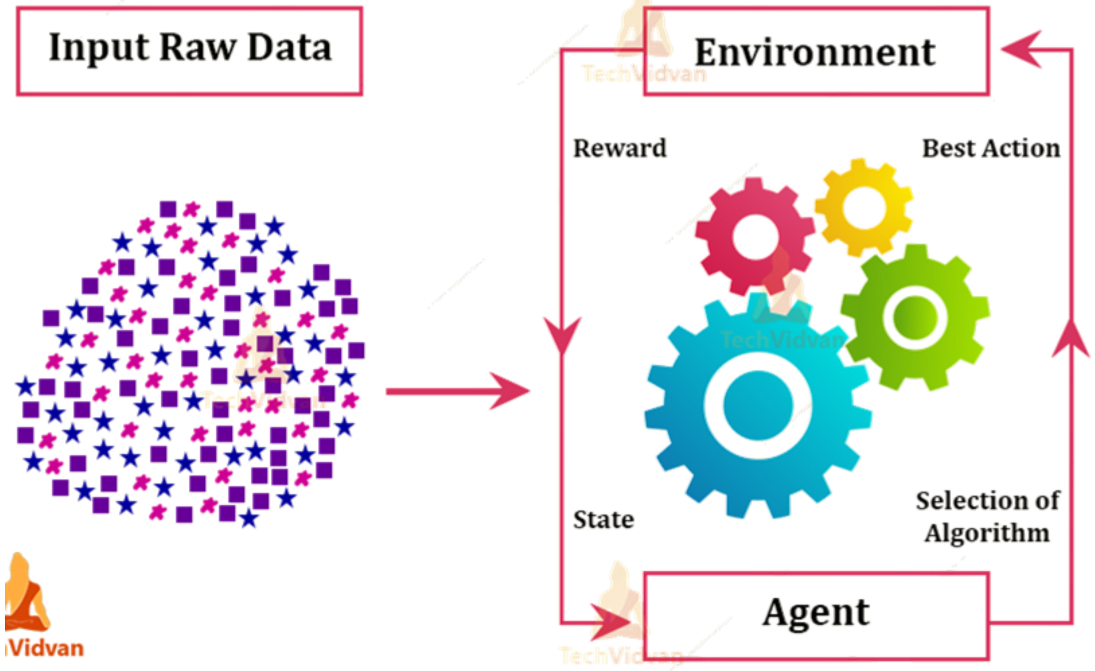


En el Aprendizaje por Refuerzo, un agente interactúa con su entorno produciendo acciones y descubriendo errores o recompensas. El Aprendizaje por Refuerzo está inspirado en la psicología conductual y consiste en tomar secuencias de decisiones con el objetivo de maximizar una recompensa acumulada.

Qué es Machine Learning?

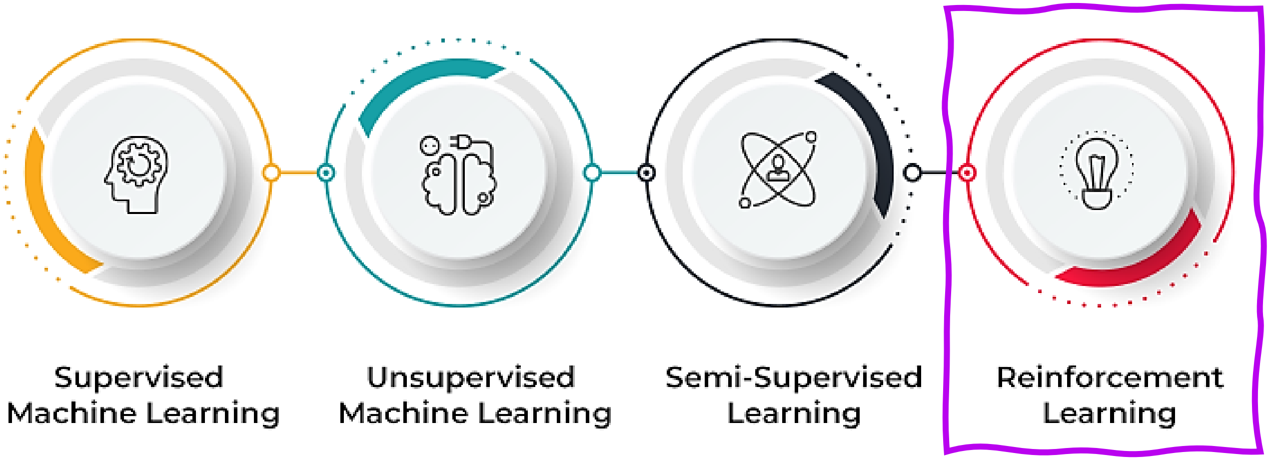


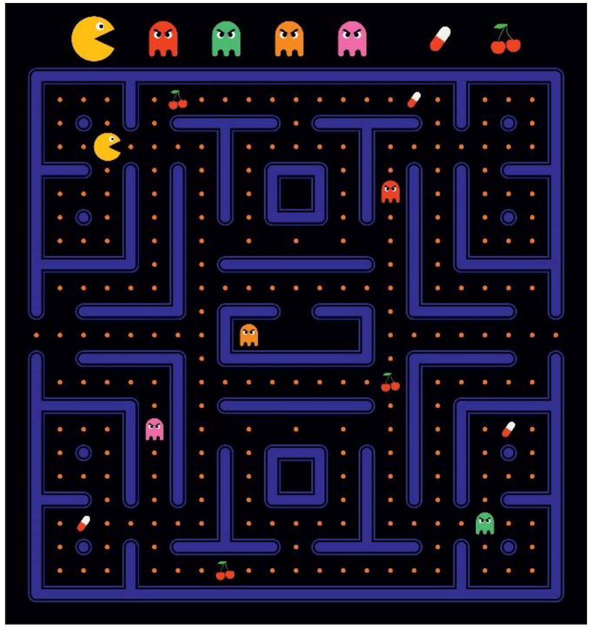




Link

Qué es Machine Learning?





Link

Agente: El aprendiz que interactúa con el entorno. El agente realiza acciones basándose en su estado actual y en sus observaciones.

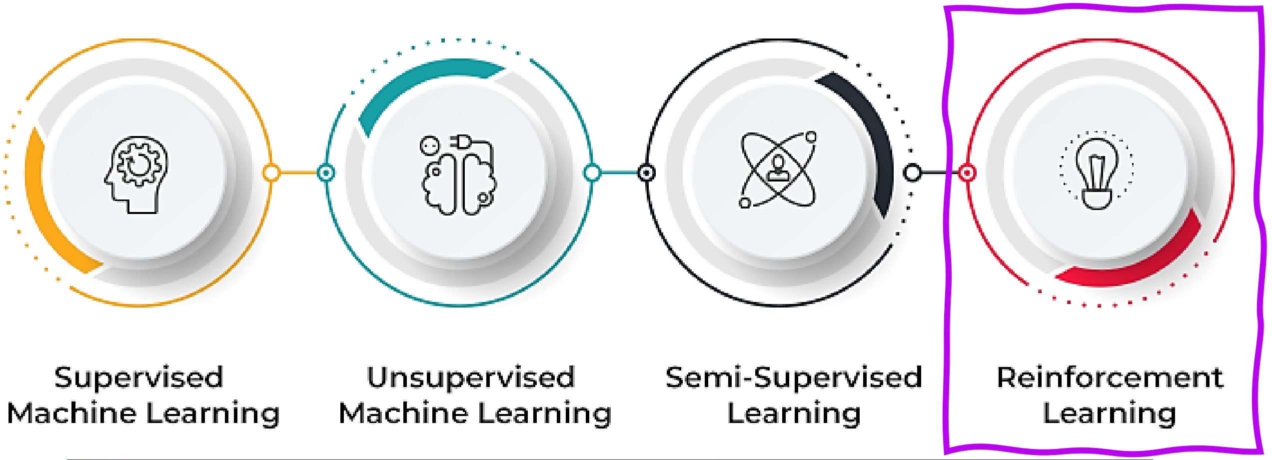
Entorno: El sistema externo con el que interactúa el agente. El entorno es responsable de proporcionar retroalimentación al agente en forma de recompensas y transiciones de estado.

Estado: Una representación de la situación o configuración actual del entorno. El estado ayuda al agente a comprender su posición y a tomar decisiones.

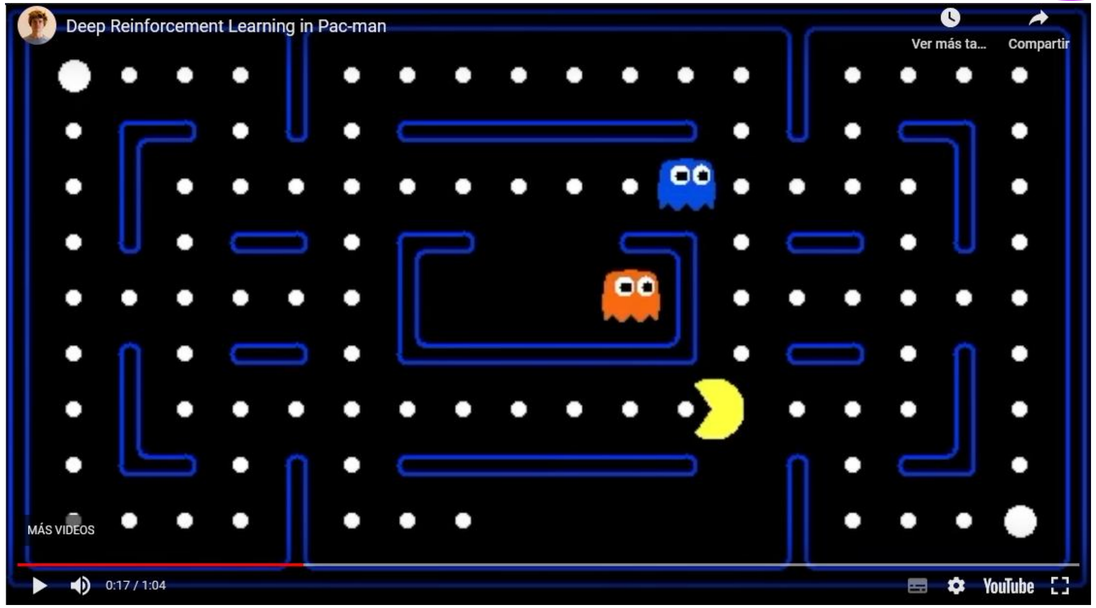
Acción: Las elecciones o decisiones que toma el agente en cada instante de tiempo. Las acciones se seleccionan a partir de un conjunto de acciones posibles disponibles para el agente en el estado actual.

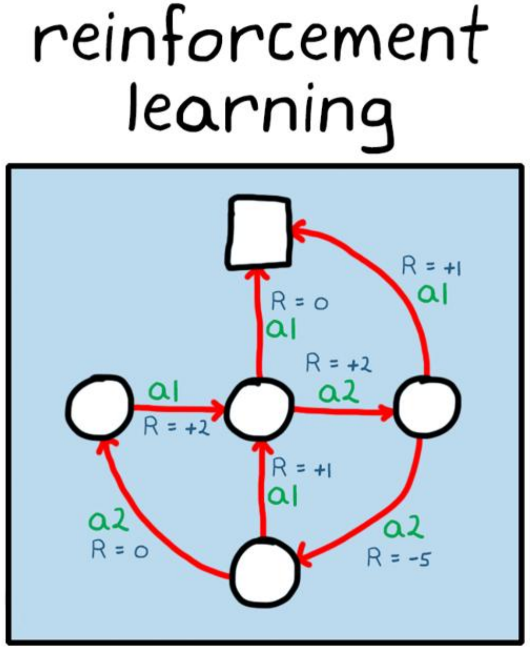
Recompensa: Un valor escalar que el agente recibe del entorno después de realizar una acción en un estado determinado.

Qué es Machine Learning?



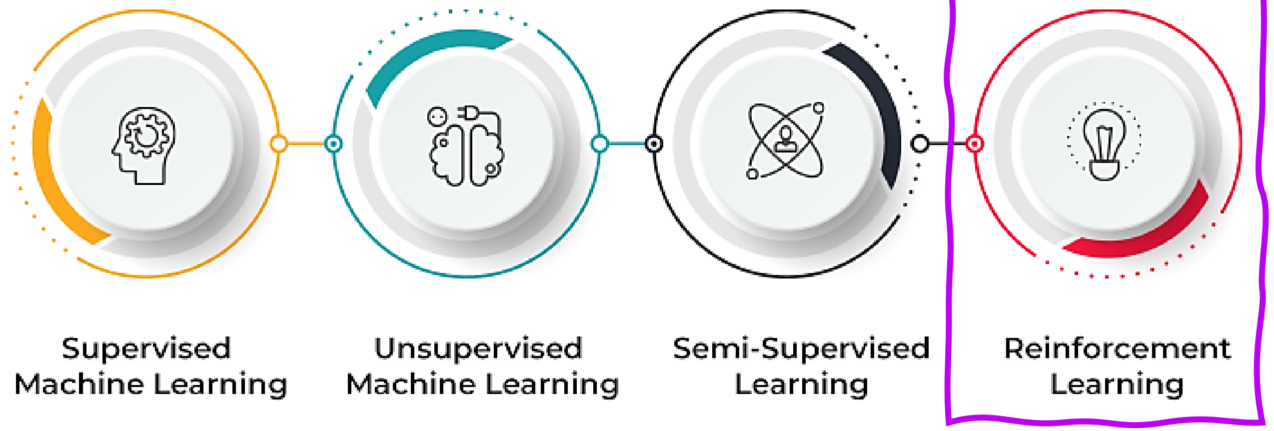
The agent determine by own what information is relevant and how to implement to learn.





https://www.youtube.com/watch?v=QilHGSYbjDQ&t=17s

Qué es Machine Learning?



Requirements:

- Data in numeric format (Hot encoding)
- Data store in array

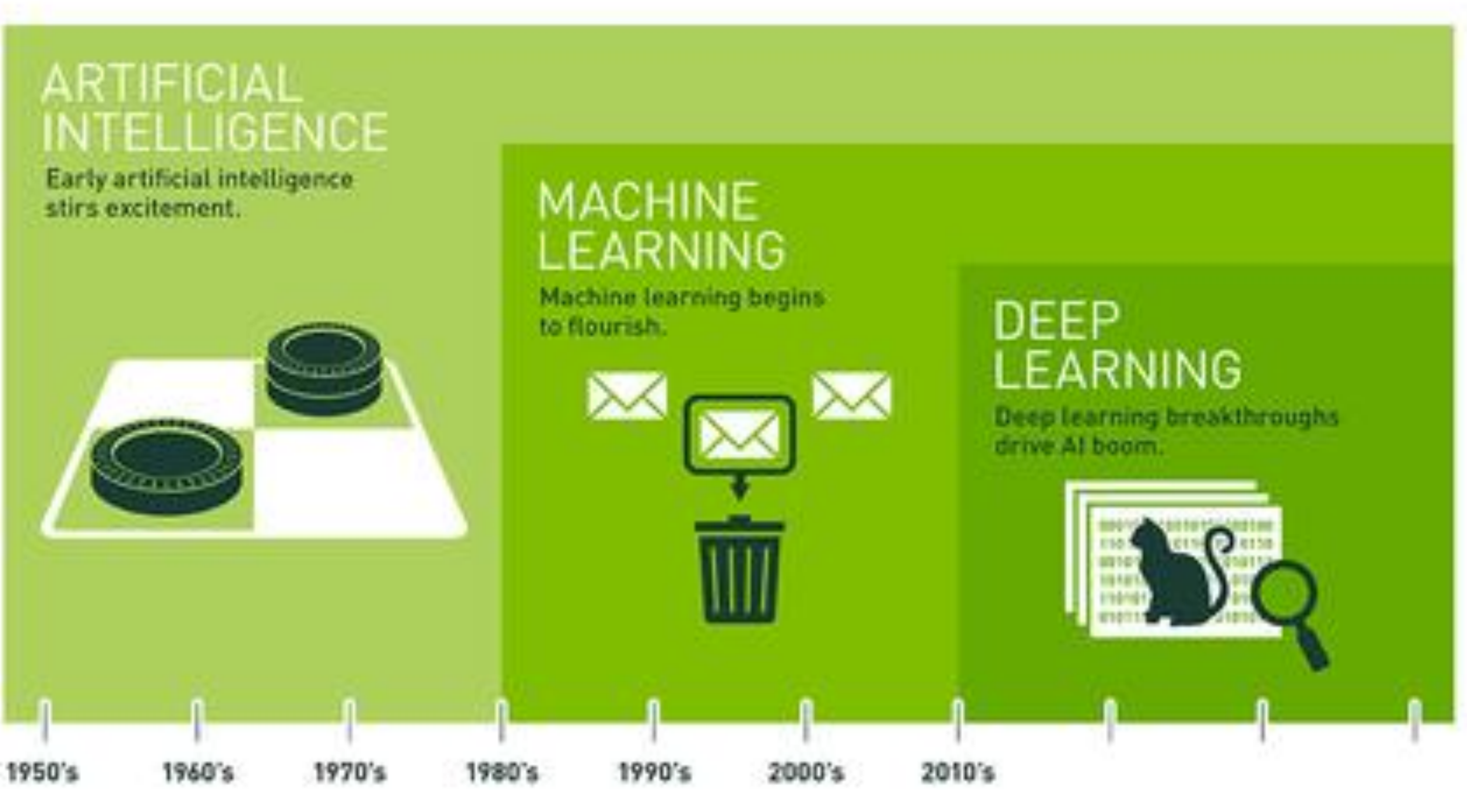
Pros:

- Reinforcement learning can be used to solve very complex problems that cannot be solved by conventional techniques
- The model can correct the errors that occurred during the training process
- In the absence of a training dataset, it is bound to learn from its experience
- Capable of finding previously unknown patterns in data
- Real time analysis

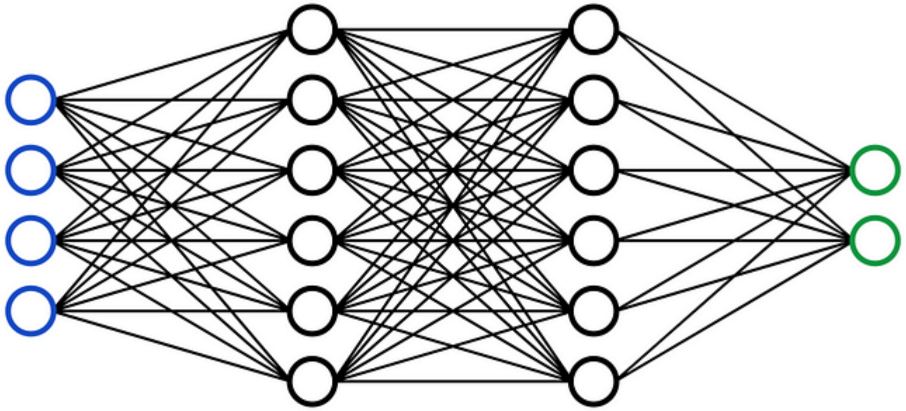
Cons:

- Demands a lot of data and a lot of computation.
- Too much of reinforcement may cause an overload which could weaken the results.

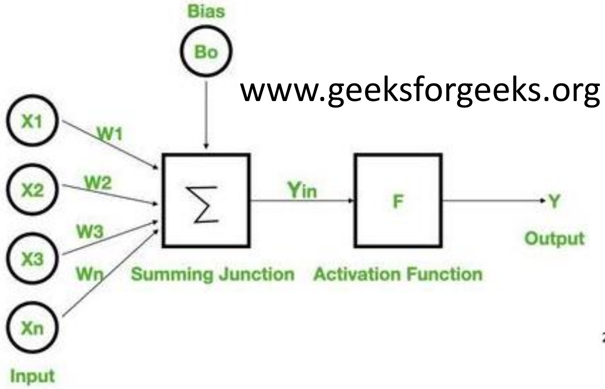
Deep Learning



Deep Learning

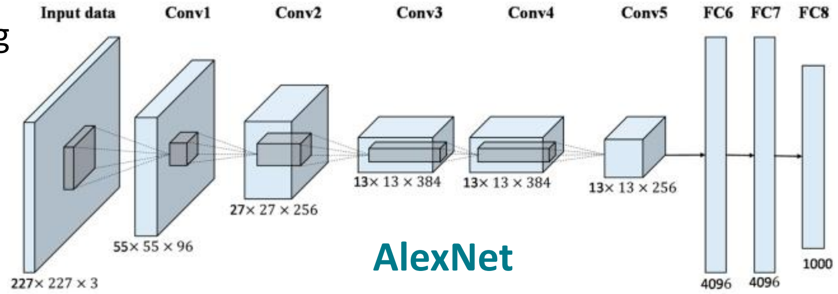


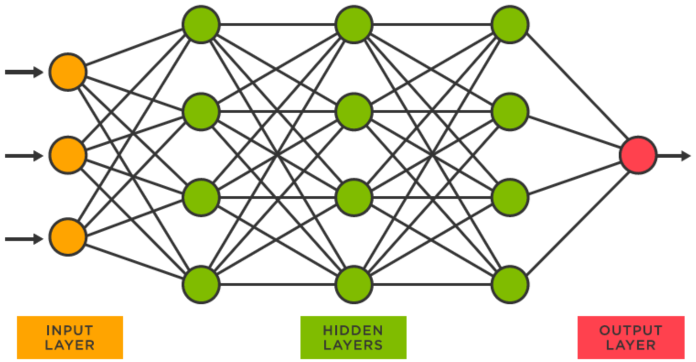
Dog
Cat



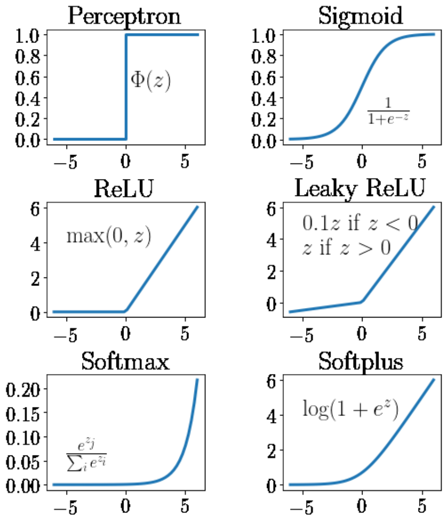
www.geeksforgeeks.org

DOI:doi.org/10.3390/rs9080848



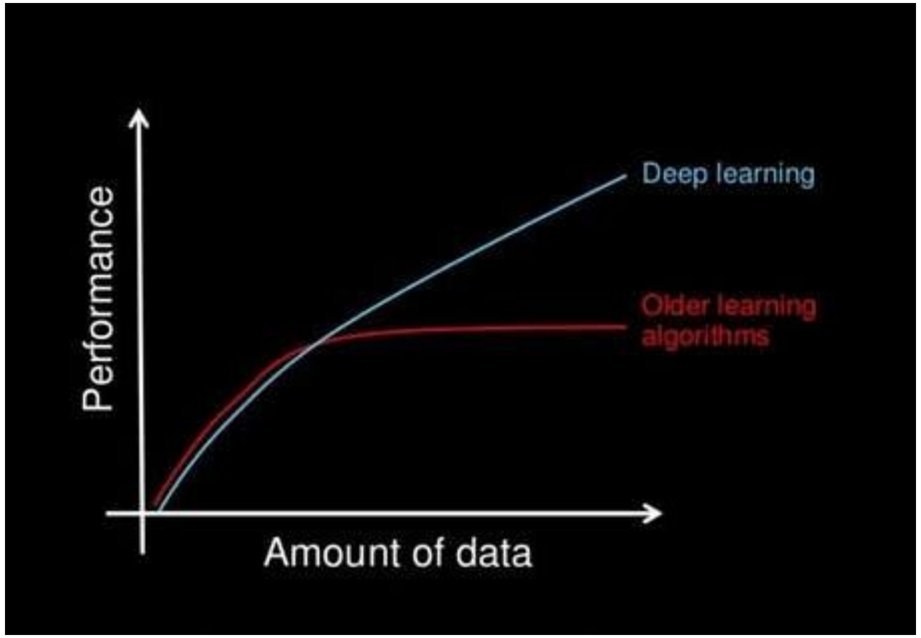


www.tibco.com



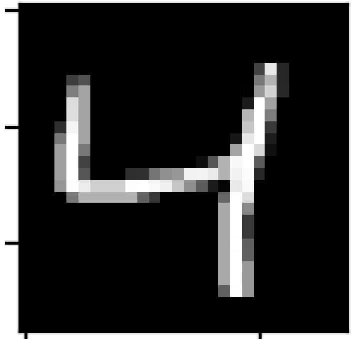
[Aaron Stebner]()

Why to use Deep Learning?

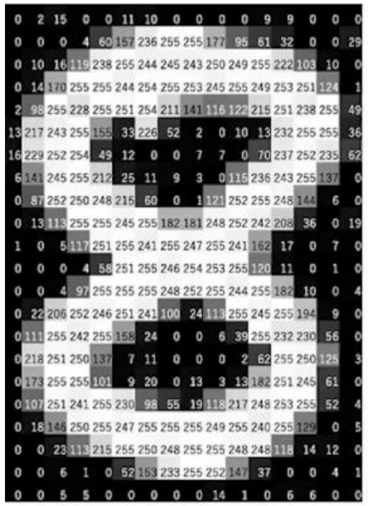


https://www.youtube.com/watch?v=O0VN0pGgBZM&t=35s

What number is this?

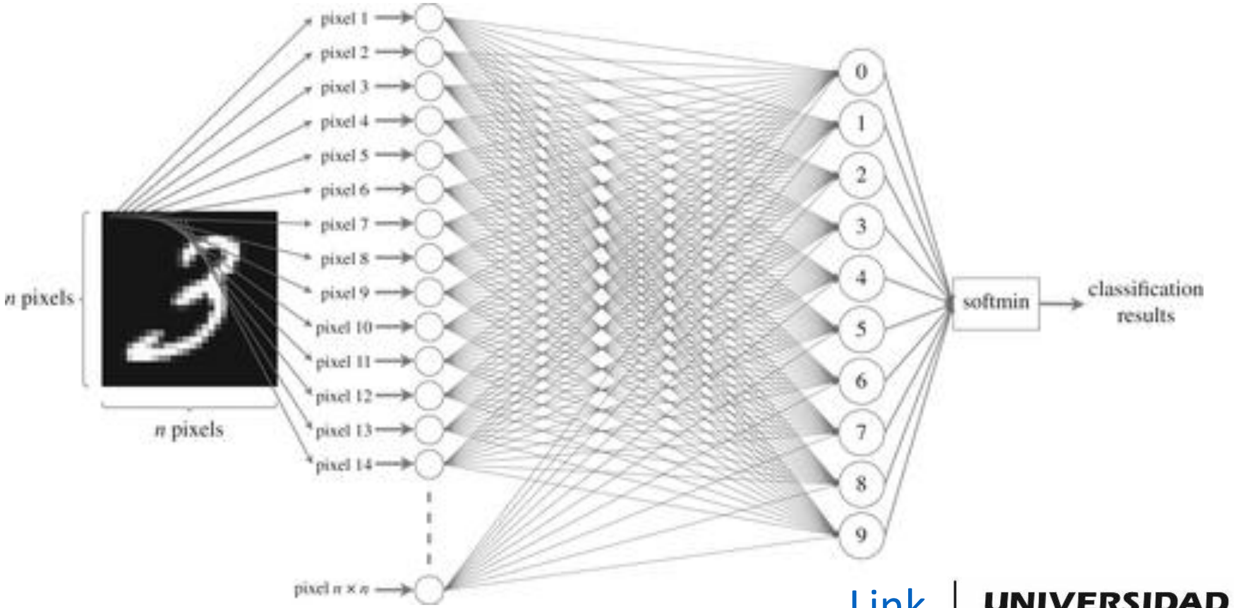


Write a code to determine what number is this



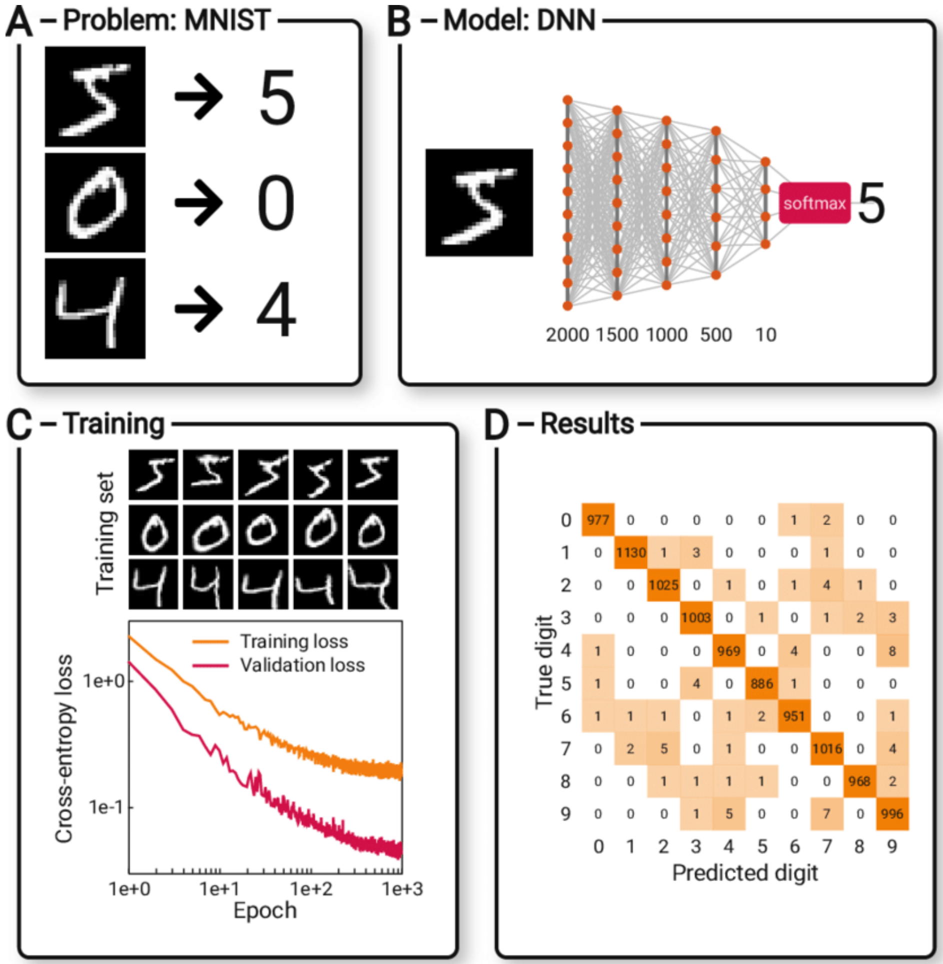
Discretizing the information by pixels

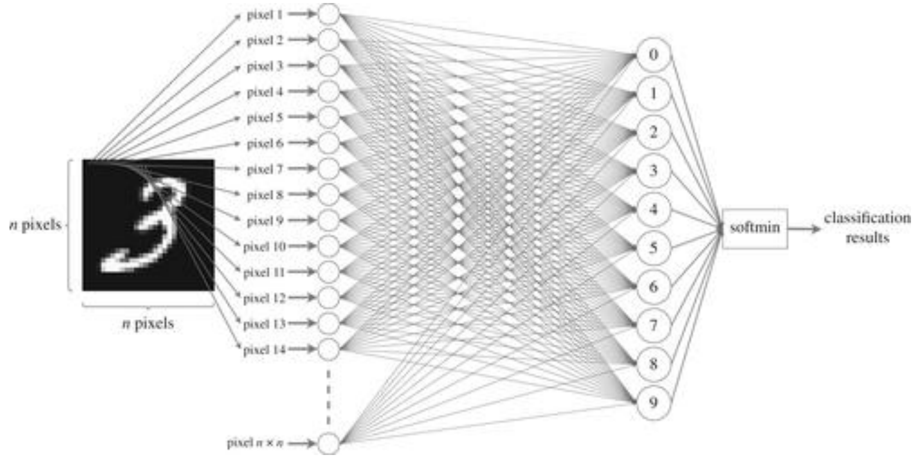
Processing all pixel together



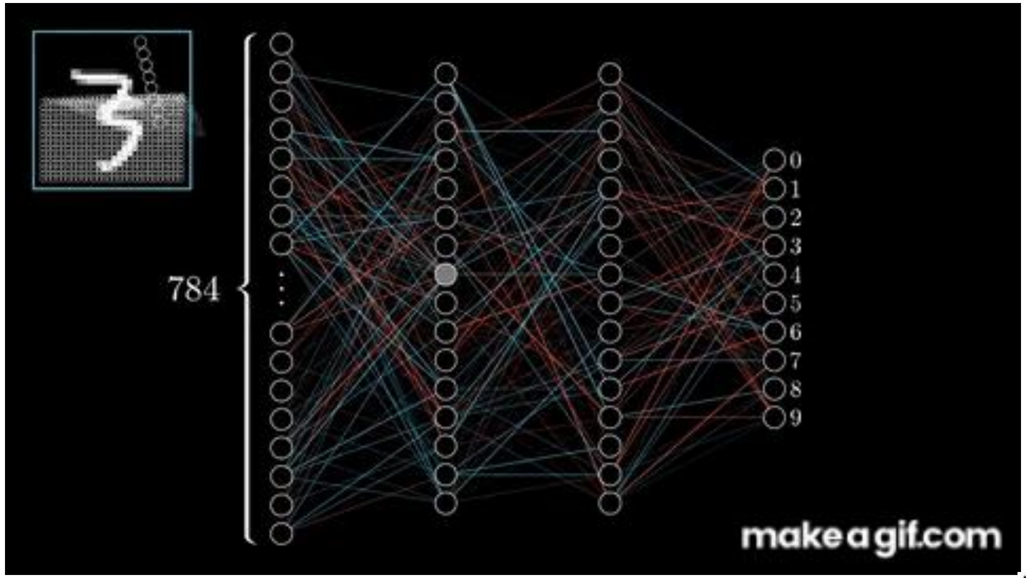
[Link](#)

Deep Learning

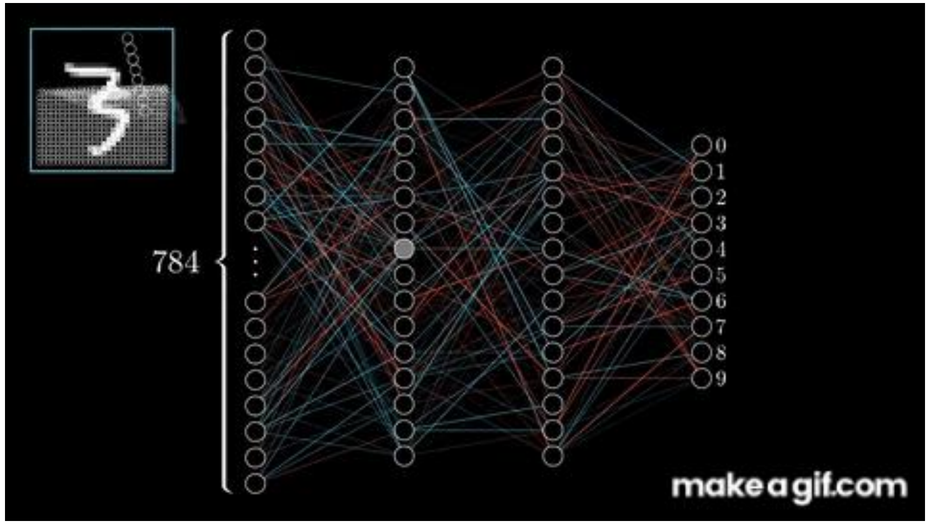


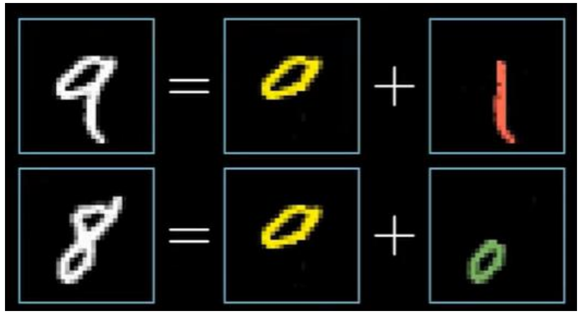


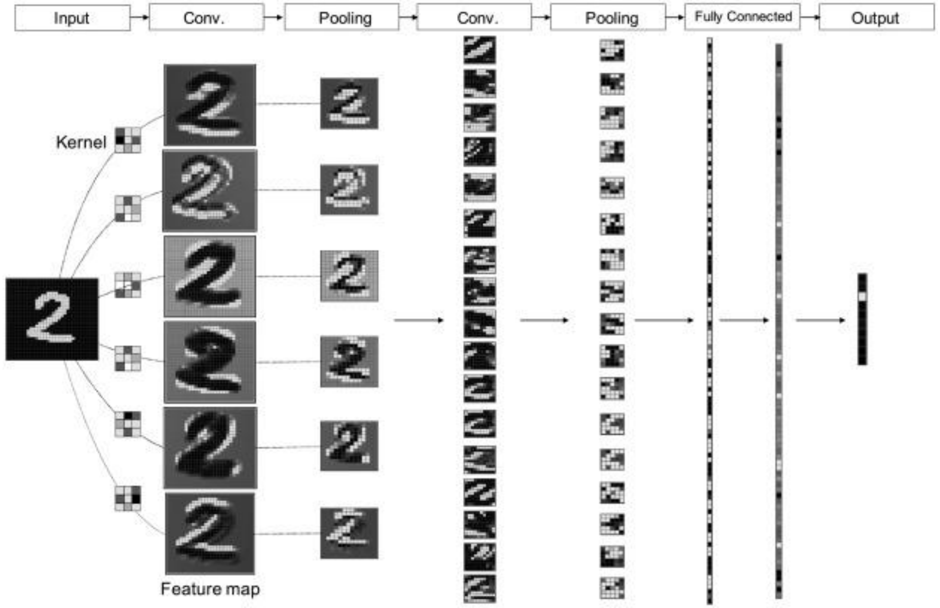
Activations in the hidden layer that follows a specific behavior



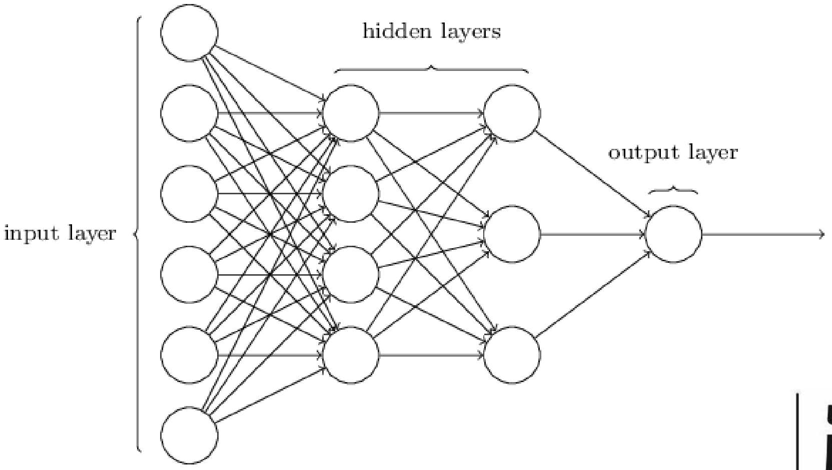
Activations in the hidden layer that follows a specific behavior







https://doi.org/10.1016/j.iatssr.2019.11.008

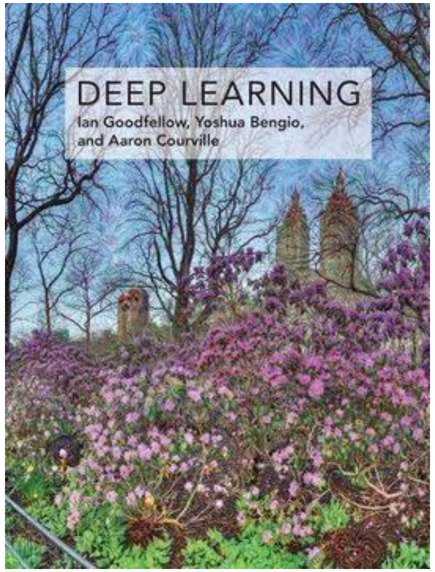


Extract the most important features

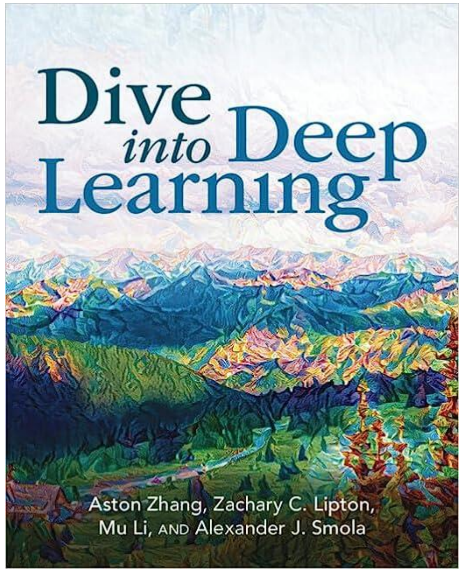
Adjust weights

Bibliography / Sources

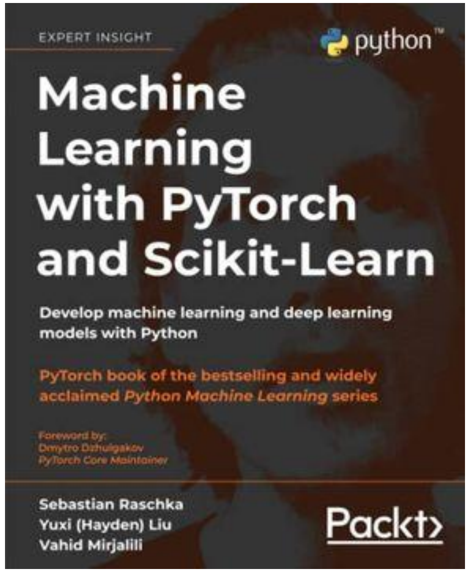
“Deep Learning"



Dive into Deep Learning



Machine Learning with Pytorch and Scikit-Learn



Resources

DeepMind

datacamp

coursera

HUGGING FACE

https://towardsdatascience.com/

https://distill.pub/

https://d2l.ai/index.html

https://neptune.ai/blog

https://medium.com/

Thank you all!